An inexact two-stepsize stochastic SQP method for nonlinear optimization problems with rank-deficient constraint Jacobians

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Abstract

In this paper, we mainly consider nonlinear optimization problems with a stochastic objective and deterministic equality constraints. We propose a stochastic sequential quadratic programming (SQP) algorithm and analyze its worst-case complexity under mild assumptions. The method utilizes a step decomposition strategy and reduces stochastic variance by assigning different stepsizes to different components of the step direction. A first known $\mathcal{O}(\epsilon_c^{-2})$ worst-case complexity of the infeasibility measure is proved when no constraint qualification is assumed. We also discuss the use of inexact iterative linear system solvers in solving the subproblems. Numerical experiments show that our algorithm has a superior infeasibility convergence performance and competitive KKT convergence rate compared to the state-of-the-art stochastic SQP method.

1 Introduction

In this paper, we consider the equality-constrained optimization problem as follows:

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{s.t.} \quad c(x) = 0, \tag{1}$$

where the objective function $f : \mathbb{R}^n \to \mathbb{R}$ and the constraints $c : \mathbb{R}^n \to \mathbb{R}^m$ are smooth.

Such problems emerge in various important applications throughout science and engineering, including optimal control [8, 33], PDE-constrained optimization [21], and network optimization [7]. Applications can also be seen in various areas of machine learning, such as transfer learning [19] and physics-informed deep learning [12, 24].

There are two types of approach that are widely used to solve problem (1). Penalty-based methods, such as the augmented Lagrangian method, convert problems into unconstrained ones and handle them with unconstrained optimization algorithms. Sequential Quadratic Programming (SQP) methods, on the other hand, solve a sequence of quadratic subproblems to compute step directions. While this paper focuses on the analysis of an SQP method, we will introduce both methods in detail in the literature review section.

Classic theories of SQP methods are built upon deterministic problems and constraints [34], while recently much attention has been paid to stochastic problems [2, 3, 25]. More specifically, our problem of interest has stochastic objective and deterministic constraints, in the sense that the objective is the expectation of some function under an unknown distribution.

Apart from introducing stochasticity, researchers have been working on relaxing the constraint qualifications. Traditionally, constraint qualifications, such as the Linear Independence Constraint Qualification (LICQ) and the Mangasarian-Fromovitz Constraint Qualification (MFCQ), are assumed for contrained optimization problems. However, the rank deficiency of constraint Jacobians is often encountered in practice, which could ruin the convergence results if the algorithm is not specifically designed. In [13], this kind of problem was discussed using trust-region SQP methods without stochasticity in the objective.

To our knowledge, few previous works have discussed stochastic SQP algorithms without constraint qualifications. [2] proposed their algorithm and proved convergence guarantees. Given the absence of complexity results on this topic, we propose and analyze a two-stepsize SQP algorithm, which utilizes inexact subproblem solvers and does not assume constraint qualifications. To analyze its performance, we use an l_2 merit function and split the behavior of the merit parameter sequence into three different cases: (i) positively bounded with no constraint qualifications; (ii) everywhere constraint qualifications; (iii) the merit parameter vanishes. A comprehensive analysis has been conducted to study the worst-case complexity results for all three cases.

We prove that our algorithm converges to stationary points (in expectation) of both infeasibility and KKT measures. Unlike some literature, the infeasibility measure we use here is $\|J_k^T c_k\|_2$, and the KKT measure is defined as $\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2$, where v_k is the normal component of the step direction. This is because we do not assume that the LICQ is valid for all iterations. To achieve $\|J_k^T c_k\|_2 \leqslant \epsilon_c$, our algorithm needs at most $\mathcal{O}(\epsilon_c^{-2})$ iterations when there is a positive lower bound on the merit parameter sequence. This is the *first known* complexity result under such mild assumptions.

Moreover, an improved $\mathcal{O}(\epsilon_c^{-1})$ worst-case complexity for the infeasibility measure is proven under the Linear Independence Constraint Qualification (LICQ). Also, we prove that our algorithm can converge to an infeasible stationary point when the merit parameter vanishes, which is the best one can hope for without a bound on the merit parameter sequence. Meanwhile, when the merit parameter sequence is bounded, a $\mathcal{O}(\epsilon_L^{-4})$ worst-case complexity of the KKT measure is obtained, which is known to be optimal for stochastic gradient methods[17].

Furthermore, we exploit the use of inexact solvers on the tangential subproblems and prove convergence of the inexact algorithm under mild assumptions on the solvers. The parallel worst-case complexity results are also proven. The numerical experiments show the enhancement in both speed and accuracy of inexact algorithms.

The paper is organized as follows. Section 2 includes a review of the literature on classic algorithms and their recent developments, especially their stochastic variants. In Section 3, our desired problem and stationarity measures are formally defined. A generic algorithm framework 3 will also be proposed. In Section 4, the worst-case complexities are proved in three different cases. Section 5 discussed the inexact solutions of the subproblems. Numerical experiments are presented in Section 6 to compare our algorithm with existing ones. Finally, Section 7 concludes our work.

1.1 Notations

The set of real numbers is denoted as $\mathbb{R}$, the set of positive real numbers is denoted as $\mathbb{R}_{>0}$, the set of natural numbers is denoted as $\mathbb{N}$, the set of n-dimensional real vectors is denoted as $\mathbb{R}^n$, the set of m-by-n dimensional real matrices is denoted as $\mathbb{R}^{m\times n}$, and the set of n-by-n dimensional symmetric matrices is denoted as $\mathbb{S}^n$. As in (1), f and c denote the objective and constraints, respectively. c_i denotes the i-th component of constraint c. Given $J \in \mathbb{R}^{m\times n}$, the null space of J is denoted as $\text{Null}(J)$, and the range space of J^T is denoted as $\text{Range}(J^T)$.

Our algorithm is iterative in the sense that, given a starting point $x_0 \in \mathbb{R}^n$, it generates a sequence of iterations $\{x_k\}$ with $x_k \in \mathbb{R}^n$ for all $k \in \mathbb{N}$. Given $J_k \in \mathbb{R}^{m\times n}$, Z_k denotes a matrix whose columns form an orthonormal basis for $\text{Null}(J_k)$.

2 Literature Review

In this section, we review the algorithm frameworks of two types of methods that are widely used in nonlinearly constrained optimization. They are penalty-based methods and sequential quadratic programming methods. We will summarize the general development of them in a *deterministic* setting. Then, we mention some recent related works under stochastic settings and compare their complexity results. We only consider the equality constrained problem1 here. For each approach, there are variants that can address inequality constraints. The reader could refer to [26] for more details and proofs.

2.1 Penalty-based methods

A higher point of view on penalty-based methods is that, instead of coping with hard constraints, we transfer the optimization problems to unconstrained ones and solve them using well-developed algorithms such as first-order methods and Newton-type methods. By adding modified constraint terms to the objective, we turn (1) into a series of unconstrained optimization problems. We mention some classical variants of penalty-based methods and provide their convergence properties as propositions without proofs.

The *quadratic penalty method* solves a sequence of unconstrained optimization problems

$$\min_{x_k} Q(x_k;\mu_k) = f(x_k) + \mu_k \sum_{i=1}^{m} c_i^2(x_k), \tag{2}$$

where the sequence of penalty parameters $\{\mu_k\}_{k>0}$ is monotonically increasing. Since we only consider equality constraints, the unconstrained problems are smooth, so they can be solved by techniques such as quasi-Newton and nonlinear conjugate gradient methods.

Proposition 2.1. *Suppose that each x_k is the exact global minimizer of $Q(x_k;\mu_k)$ defined in* (2) *and $\mu_k \uparrow \infty$. Then every limit point x^* of the sequence $\{x_k\}$ is a global solution of* (1).

However, as $\mu_k \uparrow \infty$, minimization of $Q(x_k;\mu_k)$ becomes more difficult and Hessian $\nabla_{xx}Q$ becomes arbitrarily ill-conditioned. It is practically impossible to find the exact global minimizer(s) of $Q(X_k;\mu_k)$.

Taking one step further, the *Augmented Lagrangian method* (ALM) reduces the possibility of ill-conditioning by adding extra Lagrange multiplier estimates to the objective. Specifically,

it solves the following problem:

$$\min_{x_k} L(x_k, \lambda_k; \mu_k) = f(x_k) - \sum_{i=1}^{m} \lambda_{k,i} c_i(x_k) + \frac{\mu_k}{2} \sum_{i=1}^{m} c_i^2(x_k). \tag{3}$$

The objective function can be seen either as adding an additional quadratic penalty term to the (standard) Lagrangian function, or as adding an additional term involving the dual variable λ. The KKT condition of (3) asks for $c_i(x_k) \approx -(\lambda_i^* - \lambda_{k,i})/\mu_k$. Compared with $c_i(x_k) \approx -\lambda_i^*/\mu_k$ of the quadratic penalty method, the regularized term $(\lambda_i^* - \lambda_{k,i})$ makes the infeasibility measure much smaller when the primal-dual point is close to an optimal solution. The following proposition gives the condition and error bounds of x_k and λ_k.

Proposition 2.2. *Let x^* be a local solution of (1) at which the LICQ is satisfied, and the second-order sufficient conditions are satisfied for $\lambda = \lambda^*$. Then there is a threshold value $\bar{\mu}$ such that for all $\mu > \bar{\mu}$, x^* is a strict local minimizer of $L(x, \lambda^*; \mu)$. Moreover, there exist δ, ϵ and M such that, for all λ_k and μ_k satisfying $\|\lambda_k - \lambda^*\| \leqslant \mu_k \delta$, the problem*

$$\min_x L(x, \lambda_k; \mu_k) \quad s.t.\ \|x - x^*\| \leqslant \epsilon$$

has a unique solution x_k, and $\|x_k - x^*\| \leqslant M\|\lambda^k - \lambda^*\|/\mu_k$.

This proposition illustrates some properties of ALM. Both λ_k and μ_k can affect the convergence of x_k. When our estimate of λ_k is close enough to λ^*, the quadratic penalizing parameter μ_k can be chosen relatively small. This is clearly a better strategy than the quadratic penalty method, where only μ_k can be increased. This is why ALM is preferred over the quadratic penalty method in most problems.

Then, we introduce the *nonsmooth exact penalty methods*. The unconstrained optimization problems are formulated as

$$\min_{x_k} \phi(x) = f(x_k) + \mu \sum_{i=1}^{m} \|c_i(x_k)\|, \tag{4}$$

where l_1, l_2 and l_∞ norms are commonly used. Due to nonsmoothness, the minimization of ϕ is much more difficult. However, they can be solved by local smooth models of ϕ, which can also be seen as an SQP-style technique if we take ϕ as a merit function. We will mention this later in the paper.

In practice, penalty-based methods, especially the augmented Lagrangian method, are often used for problems with linear constraints and bound constraints (in inequality constrained problems). We will discuss the complexity results of them under stochastic settings later, together with stochastic SQP methods.

2.2 SQP methods

Unlike penalty-based methods, *Sequential Quadratic Programming* methods maintain the (linearized) constraints in every subproblem. Rather than complicating the constraints by adding modified terms to the objective, SQP methods compute step directions by solving a sequence of local models of the original problem. The constraint preserving property makes SQP methods effective for solving problems with highly nonlinear constraints. In this subsection, we

mainly focus on the computation of an SQP step and discuss general line-search and trust-region frameworks.

Consider the Lagrangian function of (1):

$$L(x, y) = f(x) + c(x)^T y, \tag{5}$$

where $y \in \mathbb{R}^m$ is a vector of Lagrange multipliers. Under a constraint qualification (such as the LICQ), the first-order necessary stationary conditions with respect to (1) are given by

$$F(x, y) = \begin{bmatrix} \nabla_x L(x, y) \\ \nabla_y L(x, y) \end{bmatrix} = \begin{bmatrix} \nabla f(x) + J(x)^T y \\ c(x) \end{bmatrix} = 0. \tag{6}$$

To compute an SQP step, one way is to solve the KKT system with Newton method. The Jacobian of (6) with respect to (x, y) is given by

$$F'(x, y) = \begin{bmatrix} \nabla^2_{xx} L(x, y) & J(x)^T \\ J(x) & 0 \end{bmatrix}. \tag{7}$$

Then the Newton step update (x_k, y_k) is given by

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} + \begin{bmatrix} d_k \\ d_y \end{bmatrix}, \tag{8}$$

where (d_k, d_y) solves the system

$$\begin{bmatrix} \nabla L^2_{xx}(x_k, y_k) & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} d_k \\ d_y \end{bmatrix} = - \begin{bmatrix} \nabla f_k + J_k^T y_k \\ c_k \end{bmatrix}. \tag{9}$$

This system is well-defined and has a unique solution if the matrix F' is nonsingular.

An alternative way to compute the step is to solve the following quadratic program

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & f_k + \nabla f_k^T d + \frac{1}{2} d^T \nabla^2_{xx} L(x_k, y_k) d \\ \text{s.t.} \quad & c_k + J_k d = 0, \end{aligned} \tag{10}$$

which produces an equivalent KKT system to (9)

$$\begin{bmatrix} \nabla^2_{xx} L(x_k, y_k) & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} d_k \\ y_{k+1} \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) \\ c_k \end{bmatrix}. \tag{11}$$

Both viewpoints are useful to the step direction computation. While the Newton method point of view requires constraint qualifications to solve the next step, the quadratic programming perspective allows us to use different algorithms to solve the quadratic subproblem (10) and helps overcome possible rank deficiency in constraint Jacobians. Moreover, we could use quasi-Newton methods such as BFGS updates and reduced Hessians to flexibly choose the local quadratic models. Therefore, we will adopt the latter point of view in our remaining analysis.

After computing the step direction d_k, we find a way to update the iteration. Line search and trust-region methods are widely used to update iterations. The step update procedure can generally be formulated as $x_{k+1} = x_k + \alpha_k d_k$. Generally speaking, the line search method finds the appropriate α_k, while the trust-region method finds the appropriate d_k and sets $\alpha_k = 1$. As mentioned in the nonsmooth exact penalty method, the penalized function $\phi(x)$ in (4) can be used to determine whether a stepsize is accepted. In the SQP context, ϕ is usually written in the opposite way:

$$\phi(x, \tau) = \tau f(x) + \|c(x)\|, \tag{12}$$

where τ is called the merit parameter.

The line search SQP method reduces the stepsize α_k until a model reduction condition is satisfied. Denote $D(\phi, d)$ as the directional derivative of ϕ in the direction of d. We can use the following line search process to find an appropriate α_k. Unlike the line search approach,

Algorithm 1 A General Line Search Process

Choose parameters $\eta \in (0, 0.5), \tau \in (0, 1)$
Set $\alpha_k \leftarrow 1$.
while $\phi(x_k + \alpha_k d_k; \mu_k) > \phi(x_k; \mu_k) + \eta \alpha_k D(\phi(x_k; \mu_k), d_k)$ **do**
  Set $\alpha_k \leftarrow \tau_\alpha \alpha_k$ for some $\tau_\alpha \in (0, \tau]$;
end while

the trust-region method modifies the subproblem (10) by adding a trust-region constraint $\|d\| \leqslant \Delta_k$. Then, the ratio ρ_k between the merit function reduction (ared_k) and the model reduction (pred_k) is checked. If the ratio is beyond a preset threshold, then we reduce Δ_k and re-compute d_k. A general trust-region process is given below. Either approach has its own

Algorithm 2 A General Trust-Region Process

Choose parameters $\eta, \tau \in (0, 1)$;
Compute d_k with the extra trust-region constraint $\|d_k\| \leqslant \Delta_k$.
Compute $\rho_k = \text{ared}_k / \text{pred}_k$;
if $\rho_k > \eta$ **then**
  Set $x_{k+1} = x_k + d_k$;
  Choose $\Delta_{k+1} \geqslant \Delta_k$;
else
  Set $x_{k+1} = x_k$;
  Choose $\Delta_{k+1} \leqslant \tau \Delta_k$;
end if

advantages. In line search methods, obtaining a solution of d_k is relatively inexpensive with the help of techniques such as warm-start procedures. The directional derivatives can be used for different stepsize values, which reduces computation. The trust-region process, on the other hand, can handle ill-conditioned Hessians and rank deficiencies in constraint Jacobians better.

To conclude this subsection, we present a proposition regarding the convergence rate of local SQP methods.

Proposition 2.3. *Suppose x^* is a local solution of (1) at which the following conditions hold:*

1. *The functions f and c are twice differentiable in a neighborhood of x^* with Lipschitz continuity on second derivatives.*
2. *The LICQ holds at x^*, and hence the KKT conditions are satisfied for some multipliers y^*.*
3. *The second-order sufficient conditions hold at (x^*, y^*).*

Then, if (x_0, y_0) is sufficiently close to (x^*, y^*), the pairs (x_k, y_k) generated by the SQP method converge quadratically to (x^*, y^*).

2.3 Recent algorithm development

In this part, we will discuss the recent developments of both augmented Lagrangian methods and sequential quadratic programming methods. Specifically, we consider optimization problems with *stochastic* objectives and deterministic constraints, which are the target problems of this paper. In the end of this section, we will provide a comprehensive complexity table for the algorithms mentioned in the following.

2.3.1 Stochastic augmented Lagrangian methods

The stochastic augmented Lagrangian methods usually have some restrictions on their target problems. In [35], a stochastic augmented Lagrangian method is proposed to solve convex constrained optimization problems. [19] designed a stochastic augmented Lagrangian method to solve Domain Adaption problems. [9] combined the augmented Lagrangian method with adaptive sampling and applied it to problems with a stochastic convex objective and linearly deterministic constraints. For more general constrained optimization problems, [22] proposed a stochastic inexact augmented Lagrangian method (Stoc-iALM) with a so-called PStorm step computation subroutine.

Regarding complexity results, since augmented Lagrangian methods do not keep constraints separately, they only consider a KKT measure that includes the constraint part (analogous to our $\|\nabla f(x_k) + J_k^T y_k^{\text{true}}\|_2$). A recent MLALM algorithm proposed in [31] achieved $\mathcal{O}(\max\{\epsilon_c^{-4}, \epsilon_L^{-4}\})$ worst-case complexity for KKT measures, under their mean-squared smoothness constraint qualifications. [23] analyzed the complexity of ALM on smooth nonconvex problems and proved $\tilde{\mathcal{O}}(\max\{\epsilon_c^{-4}, \epsilon_L^{-4}\})$ complexity results under weak assumptions.

2.3.2 Stochastic SQP Methods

The first stochastic SQP method was proposed in [3], where an exact step direction is computed via the KKT system of the Lagrangian function, and the stepsize is generated by a projection procedure. Such projection introduces certain adaptivity into the algorithm while avoiding line search. In [14], a first-known worst-case complexity bound of $\mathcal{O}(\epsilon_L^{-4})$ was proved for their stochastic SQP method. This is arguably the best result one can achieve on unconstrained nonconvex problems under such a stochastic setting. The paper [1] used the adaptive sampling technique and improved the iteration complexity to $\mathcal{O}(\epsilon_L^{-2})$ in the price of a higher

sample complexity ($\mathcal{O}(\epsilon_L^{-2(1+\nu)}), \nu > 1$) than in [14]. In a recent work [28], the step direction is decomposed to normal and tangential parts, and different stepsizes were assigned to them. This two-stepsize technique helps reduce stochasticity in constraints and made a breakthrough in the $\mathcal{O}(\epsilon_c^{-1})$ complexity of the infeasibility measure. In addition, [4] accelerated the algorithm by variance reduction.

Apart from the adaptivity of stepsizes and sample sizes, efforts have been made to enhance performance by utilizing *inexact* solutions to subproblems. The work [15] solves the quadratic subproblems with iterative solvers such as Minimum Residual (MINRES) [29]. This allows stochastic SQP methods to handle large-scale problems where the subproblems are too large to form or factorize. Their numerical results also showed performance enhancements in both KKT errors and infeasibility errors compared to their exact counterparts.

Another research direction is the modification of subproblems. SQP-AL [25] used a differentiable exact augmented Lagrangian function in (3) as the merit function and yielded an almost-sure convergence result. [16] used an adaptive trust-region radius for the subproblems.

While most of the SQP methods assume the LICQ, there are much fewer works discussing the problems with possibly rank-deficient constraint Jacobians. [2] first looked into this problem setting and proved convergence guarantees for their algorithm, which is based on [3].

Some other problem settings of stochastic constrained optimization problems have also been extensively studied. [30], [32], and [5] included stochasticity by assuming bounded errors in their equality constraints. [6] assumed stochastic gradient oracles and proved high-probability convergence results. We conclude this section with a table of complexity results of different algorithms. Note that when any form of constraint qualification (CQ) holds, the KKT measure is $\|\nabla f(x_k) + J_k^T y_k\|$ and the infeasibility measure is $\|c_k\|$. Otherwise, they are considered as $\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2$ and $\|J_k^T c_k\|_2$, respectively.

Algorithm	Assumption	KKT Comp.	Infeasibility Comp.
SPD[20]	x_0 feasible, CQ	$\mathcal{O}(\epsilon_L^{-5})$	$\mathcal{O}(\epsilon_c^{-5})$
MLALM[31]	mean-squared smoothness, CQ	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-4})$
SSQP[3]	LICQ, $\tau_{\min} > 0$ known	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-2})$
SQP-AL[25]	LICQ	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-4})$
TSSQP[28]	LICQ, predefined stepsizes	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-1})$
ITSQP (this paper)	**No CQ assumed**	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-2})$
	LICQ, predefined stepsizes	$\mathcal{O}(\epsilon_L^{-4})$	$\mathcal{O}(\epsilon_c^{-1})$

Table 1: Comparison of complexity results.

3 Problem Statement and Algorithm Description

Throughout our analysis, we consider the equality-constrained optimization problem (1), where the objective function $f(x) = E[F(x, \omega)]$ is an expectation of $F : \mathbb{R}^n \times \Omega \to \mathbb{R}$, under random variable ω with associated probability space $(\Omega, \mathcal{F}, P)$, and $\mathbb{E}[\cdot]$ denotes the expectation with respect to P. $c : \mathbb{R}^n \to \mathbb{R}^m$ is the constraint function, whose Jacobian matrix

is denoted as $J : \mathbb{R}^n \to \mathbb{R}^{m\times n}$. We make the following smoothness assumptions throughout the paper.

Assumption 3.1. *Let $\chi \subset \mathbb{R}^n$ be an open convex set containing the sequence of iterations $\{x_k\}$ generated by our algorithm. The objective function $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and bounded over χ and its gradient $\nabla f : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous with constant $L > 0$ and bounded over χ. The constraint function $c : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable and bounded over χ and its Jacobian function $J := \nabla c^T : \mathbb{R}^n \to \mathbb{R}^{m\times n}$ is bounded over χ and Lipschitz continuous with constant $\Gamma > 0$ (with respect to $\|\cdot\|_2$) over χ.*

From Assumption 3.1, it follows that there exist positive real numbers $(f_{\inf}, f_{\sup}, \kappa_g, \kappa_c, \kappa_J) \in \mathbb{R}\times\mathbb{R}\times\mathbb{R}_{>0}\times\mathbb{R}_{>0}\times\mathbb{R}_{>0}$, such that

$$f_{\inf} \leqslant f(x_k) \leqslant f_{\sup}, \quad \|\nabla f(x_k)\| \leqslant \kappa_g, \quad \|c_k\|_2 \leqslant \kappa_c, \quad \text{and } \|J_k\|_2 \leqslant \kappa_J \text{ for all } k \in \mathbb{N}.$$

Note that these are common assumptions on the smoothness of functions. As for the boundedness of functions and gradient values, it is not ideal to assume them under a stochastic setting, since we can only provide convergence in expectation. It can be justified that these are extra assumptions needed in constrained optimization. The bounds on functions and gradient values are important to the boundedness of our step computations and problem-specific constants.

Revisiting the Lagrangian function (5) and the KKT conditions (6), we can see that under only Assumption 3.1, it is possible for the KKT system to degenerate due to the possible rank deficiency of the constraint Jacobian matrix J. Then (6) might not necessarily be satisfied at a solution of (1), or (1) might be infeasible itself. In the latter case, we can only hope that some other measure of constraint violation converges to zero. To handle this, we employ the l_2-norm $\varphi(x) = \|c(x)\|_2$ to measure infeasibility. Hence, we can say that a point $x \in \mathbb{R}^n$ is stationary with respect to φ if and only if either:

(a) $\varphi(x) = 0$, or

(b) $\varphi(x) \neq 0$ and $\nabla\varphi(x) = \dfrac{J(x)^T c(x)}{\|c(x)\|_2} = 0$.

In fact, we mainly consider $\|J(x)^T c(x)\|_2$ as the infeasibility measure in our analysis, since $\varphi(x) = 0$ will also lead to $\|J(x)^T c(x)\|_2 = 0$.

In the discussion of the quadratic subproblem (10), the Hessian matrix $\nabla^2_{xx} L(x_k, y_k)$ can be replaced by an appropriate estimate, denoted by H_k. We now restate the general quadratic subproblem with H_k and the gradient estimate g_k, and use it for the rest of our analysis.

$$\begin{aligned} \min_{d\in\mathbb{R}^n} \quad & f_k + g_k^T d + \frac{1}{2} d^T H_k d \\ \text{s.t.} \quad & c_k + J_k d = 0, \end{aligned} \tag{13}$$

where g_k is a stochastic estimate of the true gradient $\nabla f(x_k)$, and $H_k \in \mathbb{S}^n$ estimates the Hessian. The corresponding KKT system then becomes

$$\begin{bmatrix} H_k & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} d_k \\ y_{k+1} \end{bmatrix} = - \begin{bmatrix} g_k \\ c_k \end{bmatrix}. \tag{14}$$

Notice that in this reformulated KKT system, we later assume that the only stochasticity is caused by the gradient estimate g_k. The matrix H_k can be computed by various update strategies or chosen manually, while J_k and c_k are deterministic up to the current iterate k.

3.1 Step computation and algorithm

For a general quadratic programming problem in the form (13), there are numerous ways to solve it when J_k has full row rank, such as factorization methods and conjugate gradient methods. However, we do not assume full rankness in the Jacobians. Rather than directly solving (14), we apply a step decomposition strategy (Byrd-Omojokun method), which is commonly used to handle rank deficiency.

For each iteration, we first compute the normal component $v_k \in \text{Range}(J_k^T)$ of the search direction toward linearized constraint violation with a trust-region subproblem:

$$\min_{v\in\mathbb{R}^n} \frac{1}{2}\|c_k + J_k v\|_2^2 \quad \text{s.t.} \quad \|v\|_2 \leqslant \omega\|J_k^T c_k\|_2, \tag{15}$$

where $\omega > 0$ is a deterministic user-defined parameter to control the size of the normal component. Rather than using an adaptive trust region radius sequence $\{\Delta_k\}$ (like in [16]), we intentionally link v_k with the infeasibility measure $\nabla\varphi$. When the Jacobian J_k has full rank, the problem (15) can be solved exactly. If ω is also sufficiently large, we can solve this to optimality, which yields $c_k + J_k v_k = 0$. However, this is not the case when J_k does not have full rank.

Fortunately, we only need the normal step v_k to satisfy the Cauchy decrease condition:

$$\|c_k\|_2 - \|c_k + J_k v_k\|_2 \geqslant \epsilon_v(\|c_k\|_2 - \|c_k + \alpha_k^C J_k v_k^C\|_2), \tag{16}$$

for some $\epsilon_v \in (0, 1]$. $v_k^C = -J_k^T c_k$, and α_k^C is the solution to $\min_\alpha \frac{1}{2}\|c_k + \alpha J_k v_k^C\|^2$, subject to $\alpha \leqslant \omega$. Since this condition can be satisfied simply by choosing $v_k \leftarrow v_k^C$, the normal component can be computed at a relatively low cost. To obtain a more accurate solution, one can apply the linear conjugate gradient method with Steihaug stopping conditions to find a solution that satisfies the Cauchy decrease condition.

We can also draw an important piece of information from (15) that v_k is independent of the stochasticity at iteration k. This property lays the theoretical foundation for our superior feasibility convergence rate compared to that in [2].

After computing the normal component v_k, our algorithm computes the tangential component u_k through another constrained optimization subproblem. We start by introducing the tangential subproblem with an assumption on H_k. The following assumption assumes H_k to behave positive-definite in the null space of J_k, and is typical for the constrained optimization.

Assumption 3.2. *For all $k \in \mathbb{N}$, the matrix $H_k \in \mathbb{S}^n$ is chosen independently from g_k, the sequence $\{H_k\}$ is bounded in norm by κ_H, and there exists $\zeta \in \mathbb{R}_{>0}$, such that $u^T H_k u \geqslant \zeta\|u\|_2^2$ for all $u \in$ Null(J_k).*

The subproblem to compute the tangential component u_k is formulated as

$$\min_{u\in\mathbb{R}^n} (g_k + H_k v_k)^T u + \frac{1}{2}u^T H_k u \quad \text{s.t.} \quad J_k u = 0. \tag{17}$$

Under Assumption 3.2, u_k is unique and can be obtained by solving the Newton system

$$\begin{bmatrix} H_k & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} u_k \\ y_k \end{bmatrix} = - \begin{bmatrix} g_k + H_k v_k \\ 0 \end{bmatrix}, \tag{18}$$

while the Lagrangian multiplier y_k might have multiple solutions. An exact solution to this subproblem can be obtained through Krylov subspace methods, while inexact solutions can be obtained with iterative solvers. We will later discuss the inexact u_k in Section 5.

We also assume the gradient estimate g_k to be unbiased and to have a uniformly bounded variance.

Assumption 3.3. *For all $k \in \mathbb{N}$, the stochastic gradient estimate $g_k \in \mathbb{R}^n$ is an unbiased estimator of $\nabla f(x_k)$,* i.e.,*$\mathbb{E}_k[g_k] = \nabla f(x_k)$, where $\mathbb{E}_k[\cdot]$ denotes the conditional expectation up to iteration x_k. In addition, there exists $M \in \mathbb{R}_{>0}$ such that $\mathbb{E}_k[\|g_k - \nabla f_k\|_2^2] \leqslant M$.*

After computing v_k and u_k, a natural way to compute step d_k is simply adding them together, as in [2]. However, we rescale the tangential part u_k with some pre-defined parameter sequence β_k and then add them together to form our step direction $d_k = \beta_k u_k + v_k$ instead. The next iteration x_{k+1} is then produced by $x_{k+1} = x_k + \alpha_k d_k$. The role of β_k is crucial in our analysis. Simply put, $\{\beta_k\}$ controls the variance introduced by the stochastic gradient estimates g_k. Now, we can state our algorithm in a general framework where we do not specify how to choose β_k and α_k. There are various ways to choose parameter sequences $\{\beta_k\}$ and

Algorithm 3 A Generic Two-Stepsize Stochastic SQP Algorithm Framework

Initialize $x_0 \in \mathbb{R}^n$.
for $k = 1, 2, 3, ..., K$ **do**
  Compute normal component v_k with (15).
  Compute stochastic gradient g_k.
  Compute tangential component u_k with (17).
  Choose $\beta_k > 0$.
  Set $d_k \leftarrow \beta_k u_k + v_k$.
  Choose $\alpha_k > 0$.
  Set $x_{k+1} \leftarrow x_k + \alpha_k d_k$.
end for

$\{\alpha_k\}$. Since $\{\beta_k\}$ are meant to reduce the variance of stochastic gradient estimates, they are essentially equivalent to the stepsize in classic stochastic gradient methods. In order to obtain our convergence result, we set $\beta_k = O(1/\sqrt{K})$ where K is the total number of iterations that we plan to perform, which is a standard choice in the stochastic gradient literature. For the sake of brevity, we assign $\kappa_\beta \in \mathbb{R}_{>0}$ as an upper bound such that $\beta_k \leqslant \kappa_\beta$ for all $k \in \mathbb{N}$.

On the other hand, α_k can be chosen to be independent of K because we do not need it to control the stochastic error. Specifically, we choose α_k from an interval based on the problem parameters and β_k as follows:

$$\alpha_k \in [\nu, \nu + \theta\beta_k], \tag{19}$$

where $\nu \in \mathbb{R}_{>0}$ is later defined depending on the problem parameters and $\theta \in \mathbb{R}_{>0}$ is a user-defined constant. We can see later that the choice of ν and θ interfere with the convergence result as constants. This interval type of α_k is common in stochastic SQP methods, as in [3] and [28].

4 Convergence Analysis

To start this section, we first define some "true" variables as those computed using $\nabla f(x_k)$ rather than stochastic gradient estimates g_k. For instance,

$$u_k^{\text{true}} := \underset{u\in\mathbb{R}^n}{\operatorname{argmin}} \, (\nabla f_k + H_k v_k)^T u + \frac{1}{2} u^T H_k u \quad \text{s.t.} \quad J_k u = 0, \tag{20}$$

and can be obtained by solving the "true" system

$$\begin{bmatrix} H_k & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} u_k^{\text{true}} \\ y_k^{\text{true}} \end{bmatrix} = - \begin{bmatrix} \nabla f_k + H_k v_k \\ 0 \end{bmatrix}. \tag{21}$$

Since v_k is independent from the stochastic gradient estimate g_k, v_k^{true} is identical to v_k. Therefore, $d_k^{\text{true}} = \beta_k u_k^{\text{true}} + v_k$, and it yields a useful result:

$$J_k d_k^{\text{true}} = J_k(\beta_k u_k^{\text{true}} + v_k) = J_k v_k = J_k d_k.$$

4.1 Merit function, local model and preliminary results

To study the convergence of SQP methods, a useful tool is the merit function. Recall that we use the l-2 merit function $\phi(x,\tau) = \tau f(x) + \|c(x)\|_2$ in our analysis. We introduce a local model of the merit function, $l : \mathbb{R}^n \times \mathbb{R}_{>0} \times \mathbb{R}^n \times \mathbb{R}^n$:

$$l(x,\tau,g,d) = \tau(f(x) + g^T d) + \|c(x) + J(x)d\|_2. \tag{22}$$

The local model $l(x,\tau,g,d)$ can be seen as a Taylor approximation near the current point x. l serves as a bridge between the merit function ϕ and the infeasibility measure $\nabla\varphi$. We also define the reduction in the local model for a direction d and a gradient estimate g as

$$\begin{aligned} \Delta l(x,\tau,g,d) &:= l(x,\tau,g,0) - l(x,\tau,g,d) \\ &= -\tau g^T d + \|c(x)\|_2 - \|c(x) + J(x)d\|_2. \end{aligned} \tag{23}$$

Although Algorithm 3 does not involve any computation of τ, it still plays an important role in convergence analysis. Similarly to [2], we assume that the sequence of merit parameters can be generated by the following scheme. The only difference is that we substitute values involving stochastic gradients with "true" values and β_k.

For some fixed $\sigma \in (0,1)$ and $\varepsilon_\tau \in [0,1)$, let

$$\tau_k^{trial} \leftarrow \begin{cases} \infty & \text{if } \nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} \leqslant 0 \\ \dfrac{(1-\sigma)(\|c_k\|_2 - \|c_k + J_k d_k^{\text{true}}\|_2)}{\nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}}} & \text{otherwise,} \end{cases} \tag{24}$$

and

$$\tau_k \leftarrow \begin{cases} \tau_{k-1} & \text{if } \tau_{k-1} \leqslant \tau_k^{\text{trial}} \\ \min\{(1-\varepsilon_\tau)\tau_{k-1}, \tau_k^{\text{trial}}\} & \text{otherwise.} \end{cases}$$

Hence, the merit parameter sequence $\{\tau_k\}$ is monotonically non-increasing and satisfies

$$\tau_k(\nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}}) \leqslant (1-\sigma)(\|c_k\|_2 - \|c_k + J_k d_k\|_2) \tag{25}$$

as long as $\nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} > 0$. From the fact that $\Delta l(x_k, \tau_k, \nabla f(x_k), d_k^{\text{true}}) = -\tau_k \nabla f_k^T d_k^{\text{true}} + (\|c_k\|_2 - \|c_k + J_k d_k^{\text{true}}\|_2)$ and $\sigma < 1$, we have the model reduction inequality

$$\Delta l(x_k, \tau_k, \nabla f(x_k), d_k^{\text{true}}) \geqslant \tau_k \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} + \sigma(\|c_k\|_2 - \|c_k + J_k d_k\|_2). \tag{26}$$

Now we state some useful bounds derived from the subproblems (15) and (17) used in the step computation. They link the model reduction with our infeasibility measure. Notice that these lemmas *do not* depend on the behavior of $\{\tau_k\}$. The first lemma provides a tight bound on the infeasibility measure reduction.

Lemma 4.1. *For any one dimensional optimization problem*

$$\min_{z \in \mathbb{R}^n} \Phi(z) = \frac{1}{2} a z^2 - bz \quad s.t. \quad z \leqslant \omega,$$

where $b > 0$ and $\omega > 0$, its optimal value Φ^* satisfies.

$$\Phi^* \leqslant -\frac{b}{2} \min\Big\{ \frac{b}{|a|}, \omega \Big\}.$$

Proof. See Lemma 2.1 in [11]. ■

We know that α_k^C is the solution to the problem $\min_{\alpha > 0} \frac{1}{2}\|c_k + \alpha J_k v_k^C\|^2$ with the trust-region constraint $\alpha \leqslant \omega$. Using Lemma 4.1, we have $a = \|J_k J_k^T c_k\|_2^2, b = \|J_k^T c_k\|_2^2$, and

$$\Phi^*(\alpha) = \frac{1}{2}\|J_k J_k^T c_k\|_2^2 (\alpha_k^C)^2 - \|J_k^T c_k\|_2^2 (\alpha_k^C) \leqslant -\frac{\|J_k^T c_k\|_2^2}{2} \min\Big\{ \frac{\|J_k^T c_k\|^2}{\|J_k J_k^T c_k\|_2^2}, \omega \Big\}. \tag{27}$$

Lemma 4.2. *There exists $\kappa_v \in \mathbb{R}_{>0}$ such that, for all $k \in \mathbb{N}$ with $\|c_k\|_2 > 0$,*

$$\|c_k\|_2 (\|c_k\|_2 - \|c_k + J_k v_k\|_2) \geqslant \kappa_v \|J_k^T c_k\|_2^2.$$

Proof. We first consider the Cauchy step $v_k^C = -J_k^T c_k$ of the subproblem (15). Using the previous conclusion (27), we have

$$\begin{aligned}
\frac{1}{2}(\|c_k + \alpha_k^C J_k v_k^C\|_2^2 - \|c_k\|_2^2) &= \frac{1}{2}(2\alpha_k^C c_k^T J_k v_k^C + (\alpha_k^C)^2 \|J_k v_k^C\|_2^2) \\
&= -\alpha_k^C \|J_k^T c_k\|_2^2 + \frac{1}{2}(\alpha_k^C)^2 \|J_k J_k^T c_k\|_2^2 \\
&\leqslant -\frac{1}{2}\|J_k^T c_k\|_2^2 \min\Big\{ \frac{\|J_k^T c_k\|^2}{\|J_k J_k^T c_k\|_2^2}, \omega \Big\} \\
&\leqslant -\frac{1}{2}\|J_k^T c_k\|_2^2 \min\Big\{ \frac{1}{\|J_k^T J_k\|_2}, \omega \Big\}.
\end{aligned} \tag{28}$$

Now, since v_k satisfies the Cauchy decreasing condition (16), we have the following inequality with the fact $x(x-y) \geqslant \frac{1}{2}(x^2 - y^2)$ for any $x, y \in \mathbb{R}$ that

$$\begin{aligned}
\|c_k\|_2 (\|c_k\|_2 - \|c_k + J_k v_k\|_2) &\geqslant \epsilon_v \|c_k\|_2 (\|c_k\|_2 - \|c_k + \alpha_k^C J_k v_k^C\|_2) \\
&\geqslant \frac{1}{2}\epsilon_v (\|c_k\|_2^2 - \|c_k + \alpha_k^C J_k v_k^C\|_2^2) \\
&\geqslant \frac{\epsilon_v}{2}\|J_k^T c_k\|_2^2 \min\{\omega, \frac{\omega^2}{\|J_k^T J_k\|_2}\}.
\end{aligned} \tag{29}$$

Since $\|J_k^T J_k\|_2$ is bounded from above by Assumption 3.1, there exists a constant $0 < \kappa_v < \min\{\omega, \frac{\omega^2}{\|J_k^T J_k\|_2}\}$ such that

$$\|c_k\|_2(\|c_k\|_2 - \|c_k + J_k v_k\|_2) \geqslant \kappa_v \|J_k^T c_k\|_2^2.$$

■

Lemma 4.3. *There exists $\underline{\omega} \in \mathbb{R}_{>0}$ such that, for all $k \in \mathbb{N}$ with $\|c_k\|_2 > 0$,*

$$\underline{\omega}\|J_k^T c_k\|_2^2 \leqslant \|v_k\|_2 \leqslant \omega\|J_k^T c_k\|_2.$$

Proof. The right inequality comes from the trust-region constraint of (15). From triangle inequality and Cauchy-Schwarz inequality of norms, we have

$$\|c_k\|_2 - \|c_k + J_k v_k\|_2 \leqslant \|J_k v_k\|_2 \leqslant \|J_k\|_2 \|v_k\|_2.$$

If $\|J_k\|_2 = 0$, the desired inequality is trivial. Otherwise, by Lemma 4.2, we have

$$\|v_k\|_2 \geqslant \frac{\|c_k\|_2 - \|c_k + J_k v_k\|}{\|J_k\|_2} \geqslant \frac{\kappa_v \|J_k^T c_k\|^2}{\|J_k\|_2 \|c_k\|_2}.$$

Since J_k and c_k are assumed to be bounded in norm, there exists some $\underline{\omega} > 0$ that satisfy $\|v_k\|_2 \geqslant \underline{\omega}\|J_k^T c_k\|_2^2$ for all k. ■

Lemmas 4.2 and 4.3 are useful in the Byrd-Omojokun type of methods [27]. While their proofs do not need constraint qualifications, stronger versions can be derived under the LICQ, which are discussed later in Section 4.3.

Lemma 4.4. *For all $k \in \mathbb{N}$, the norm of u_k^{true} is bounded.*

$$\|u_k^{true}\|_2 \leqslant (\kappa_g + \kappa_H \omega \kappa_J \kappa_c)/\zeta =: \kappa_u.$$

Proof. From (21), we know $H_k u_k^{\text{true}} + J_k^T y_k^{\text{true}} = -\nabla f(x_k) + H_k v_k$. By Assumption 3.2, we have

$$\begin{aligned}
\zeta\|u_k^{\text{true}}\|_2^2 &\leqslant (u_k^{\text{true}})^T H_k u_k^{\text{true}} \\
&= -\nabla f(x_k)^T u_k^{\text{true}} - v_k^T H_k u_k^{\text{true}} \\
&\leqslant \|\nabla f(x_k)\|_2 \|u_k^{\text{true}}\|_2 + \kappa_H \|v_k\|_2 \|u_k^{\text{true}}\|_2 \\
&\leqslant \|u_k^{\text{true}}\|_2(\kappa_g + \kappa_H \omega \|J_k^T c_k\|_2) \\
&\leqslant \|u_k^{\text{true}}\|_2(\kappa_g + \kappa_H \omega \kappa_J \kappa_c).
\end{aligned}$$

Dividing both sides by $\zeta\|u_k^{\text{true}}\|_2$ yields the result. ■

The following lemma bounds the distance between the computed direction d_k and the true direction d_k^{true}, which reflects the variance of the stochastic estimates.

Lemma 4.5. *For any $k \in \mathbb{N}$, $\mathbb{E}_k[u_k] = u_k^{true}$, $\mathbb{E}_k[d_k] = d_k^{true}$, and $\mathbb{E}_k[\|d_k - d_k^{true}\|_2] \leqslant \beta_k \zeta^{-1}\sqrt{M}$.*

Proof. Under assumption 3.2, there exists a matrix Z_k whose columns form an orthogonal basis for Null(J_k), and vector w_k, w_k^{true} such that $u_k = Z_k w_k$ and $u_k^{\text{true}} = Z_k w_k^{\text{true}}$. From the subproblem (17), we can actually have an explicit form of w_k and w_k^{true}:

$$w_k = -(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k + H_k v_k); \quad w_k^{\text{true}} = -(Z_k^T H_k Z_k)^{-1} Z_k^T (\nabla f(x_k) + H_k v_k).$$

Since $(Z_k^T H_k Z_k)^{-1} Z_k^T$ and Z_k are both linear operators, from Assumption 3.3 and the property of expectation, we have $\mathbb{E}_k[u_k] = u_k^{\text{true}}$. Also, $\mathbb{E}_k[d_k] = \mathbb{E}_k[\beta_k u_k + v_k] = \beta_k \mathbb{E}_k[u_k] + v_k = \beta_k u_k^{\text{true}} + v_k = d_k^{\text{true}}$. The rest of the lemma follows from Jensen's inequality and Assumptions 3.3 and 3.2.

$$\begin{aligned}
\mathbb{E}_k[\|d_k - d_k^{\text{true}}\|_2] = \mathbb{E}_k[\|\beta_k(u_k - u_k^{\text{true}})\|_2] &= \beta_k \mathbb{E}_k[\|Z_k(w_k - w_k^{\text{true}})\|_2] \\
&= \beta_k \mathbb{E}_k[\|Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k - \nabla f(x_k)\|_2] \\
&\leqslant \beta_k \mathbb{E}_k[\|Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T\|_2 \|g_k - \nabla f(x_k)\|_2] \\
&= \beta_k \|(Z_k^T H_k Z_k)^{-1}\|_2 \mathbb{E}_k[\|g_k - \nabla f(x_k)\|_2] \\
&\leqslant \beta_k \zeta^{-1} \sqrt{\mathbb{E}_k[\|g_k - \nabla f(x_k)\|_2^2]} \\
&\leqslant \beta_k \zeta^{-1} \sqrt{M}.
\end{aligned}$$

■

Lemma 4.6. *For all $k \in \mathbb{N}$, $\mathbb{E}_k[\|u_k\|_2^2] \leqslant \zeta^{-2} M + \zeta^{-1} (u_k^{true})^T H_k u_k^{true}$.*

Proof. Since $u_k \in \text{Null}(J_k)$, by Assumption 3.2, we have

$$\begin{aligned}
\mathbb{E}_k\|u_k\|_2^2 - \zeta^{-1}(u_k^{\text{true}})^T H_k u_k^{\text{true}} &\leqslant \zeta^{-1}(\mathbb{E}_k[u_k^T H_k u_k] - (u_k^{\text{true}})^T H_k u_k^{\text{true}}) \\
&= \zeta^{-1}(\nabla f(x_k)^T u_k - \mathbb{E}_k[g_k^T u_k]) \\
&\leqslant \zeta^{-2}(-\|Z_k^T g_k\|_2^2 + \mathbb{E}_k[\|Z_k^T g_k\|_2^2]) \\
&\leqslant \zeta^{-2} M.
\end{aligned} \tag{30}$$

■

Lemma 4.6 proves an important upper bound on the expectation of $\|u_k\|_2^2$. Combining it with our next lemma, they bridge the reduced KKT measure with $\mathbb{E}\|u_k\|_2^2$.

Lemma 4.7. *For all $k \in \mathbb{N}$, it holds that $\kappa_H^{-2} \zeta \|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2^2 \leqslant (u_k^{true})^T H_k u_k^{true}$.*

Proof. From the proof of Lemma 4.5, we know that

$$\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2 = \|(Z_k^T H_k Z_k) w_k^{\text{true}}\|_2 \leqslant \kappa_H \|u_k^{\text{true}}\|_2.$$

From Assumption refass3, we know $\|u_k^{\text{true}}\|_2^2 \leqslant \zeta^{-1}(u_k^{\text{true}})^T H_k u_k^{\text{true}}$, and we have

$$\kappa_H^{-2} \zeta \|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2^2 \leqslant (u_k^{\text{true}})^T H_k u_k^{\text{true}}.$$

■

Since the sequence $\{\tau_k\}$ is monotonically non-increasing and non-negative, by the monotone convergence theorem, there are two possible tail behaviors:

(i) There exists some $\tau_{\min} \in \mathbb{R}_{>0}$ such that $\lim_{k\to\infty} \tau_k = \tau_{\min}$,

(ii) $\{\tau_k\}$ vanishes, *i.e.,* $\lim_{k\to\infty} \tau_k = 0$.

In event (i), since $\tau_{\min} \leqslant \tau_k$ for all $k \in \mathbb{N}$, by (26) we know that for all $k \in \mathbb{N}$,

$$\Delta l(x_k, \tau_{\min}, \nabla f(x_k), d_k^{\text{true}}) \geqslant \tau_{\min}\beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} + \sigma(\|c_k\|_2 - \|c_k + J_k d_k\|_2). \tag{31}$$

This generic bound on the model reduction frees us from computing τ_k values, thus avoiding the over-complication of the analysis and extra computation of τ_k at each iteration, unlike in [2] where τ_k needs to be computed using g_k.

Event (ii) leads the merit function ϕ to ignore the objective f in a limit sense and to turn to optimizing the infeasibility measure $\|J_k^T c_k\|_2$ only, which is the best we can hope for when $\{J_k\}$ is not assumed to have full rank.

We will discuss the convergence results under different events. In fact, event (i) could happen with or without full row rank of J_k, which would lead to different complexity results of the infeasibility measure. Therefore, we will mainly discuss *three* cases in the following subsections.

4.2 Case I: $\tau_{\min} > 0$ and no constraint qualifications

In this subsection, we assume that $\tau_k \searrow \tau_{\min} > 0$ and no additional constraint qualification is satisfied. We first state a generic descent lemma that relates the improvement of the merit function to the local model. This lemma only requires the existence of $\tau_{\min} > 0$.

Lemma 4.8. *Let Assumptions* 3.1*,* 3.3*, and* 3.2 *hold, and* α_k *is chosen as in* (19)*. Then if there exist* $\tau_{\min} > 0$ *and* $\alpha_k \leqslant 1$*, we have*

$$\begin{aligned}\phi(x_k + \alpha_k d_k, \tau_{\min}) - \phi(x_k, \tau_{\min}) \leqslant -\alpha_k \Delta l(x_k, \tau_{\min}, \nabla f(x_k), d_k^{true}) + \frac{\alpha_k^2 \beta_k^2}{2}(\tau_{\min} L + \Gamma)\|u_k\|_2^2 \\ + \frac{\alpha_k^2}{2}(\tau_{\min} L + \Gamma)\|v_k\|_2^2 + \alpha_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{true}).\end{aligned}$$

Proof. By the L-Lipschitz continuity of $\nabla f(x)$ and Γ-Lipschitz continuity of J_k, we have

$$\begin{aligned}&\phi(x_k + \alpha_k d_k, \tau_{\min}) - \phi(x_k, \tau_{\min}) \\ &\quad = \tau_{\min}[f(x + \alpha_k d_k) - f(x_k)] + \|c(x_k + \alpha_k d_k)\|_2 - \|c(x_k)\|_2 \\ &\quad \leqslant \alpha_k \tau_{\min} \nabla f(x_k)^T d_k + \alpha_k \|c_k + J_k d_k\|_2 + |1 - \alpha_k| \|c_k\|_2 - \|c_k\|_2 + \frac{1}{2}(\tau_{\min} L + \Gamma)\alpha_k^2 \|d_k\|_2^2 \\ &\quad = \alpha_k \tau_{\min} \nabla f(x_k)^T d_k - \alpha_k(\|c_k\|_2 - \|c_k + J_k d_k\|_2) + \frac{1}{2}(\tau_{\min} L + \Gamma)\alpha_k^2 \|d_k\|_2^2 \end{aligned} \tag{32}$$

Since $d_k = \beta_k u_k + v_k$ and u_k, v_k are orthogonal, we have $\|d_k\|_2^2 = \beta_k^2 \|u_k\|_2^2 + \|v_k\|_2^2$. Combining

with $J_k d_k = J_k d_k^{\text{true}}$, we have

$$\begin{aligned}
&\phi(x_k + \alpha_k d_k, \tau_{\min}) - \phi(x_k, \tau_{\min}) \\
&\leqslant \alpha_k \tau_{\min} \nabla f(x_k)^T d_k^{\text{true}} - \alpha_k (\|c_k\|_2 - \|c_k + J_k d_k^{\text{true}}\|_2) \\
&\quad + \frac{\alpha_k^2}{2}(\tau_{\min} L + \Gamma)(\beta_k^2 \|u_k\|_2^2 + \|v_k\|_2^2) + \alpha_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{\text{true}}) \\
&\leqslant -\alpha_k \Delta l(x_k, \tau_{\min}, \nabla f(x_k), d_k^{\text{true}}) + \frac{\alpha_k^2 \beta_k^2}{2}(\tau_{\min} L + \Gamma)\|u_k\|_2^2 + \frac{\alpha_k^2}{2}(\tau_{\min} L + \Gamma)\|v_k\|_2^2 \\
&\quad + \alpha_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{\text{true}}).
\end{aligned} \tag{33}$$

■

Lemma 4.8 relates the reduction in merit function ϕ to the local model l, norms of v_k and u_k, and the stochastic error in d_k.

Recall that α_k is chosen from the interval $[\nu, \nu + \theta\beta_k]$. Now we choose

$$\nu \in (0, \min\{\frac{\sigma \kappa_v \kappa_c^{-1}}{2(\tau_{\min} L + \Gamma)\omega^2}, 1 - \theta\kappa_\beta\}],$$

which ensures $\alpha_k \leqslant 1$. However, a drawback of this step size scheme is that we must have knowledge of the Lipschitz constants Γ, L, and the bound of the merit parameter $\tau_{\min}$, which seems unreasonable in practice. Practically, one could consider adaptive parameter choosing schemes such as the AdaGrad type in [28], which will yield similar complexity results with extra logarithmic terms.

Lemma 4.9. *Let Assumptions* 3.1, 3.3, *and* 3.2 *hold, then*

$$\mathbb{E}_k[\alpha_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{true})] \leqslant \beta_k^2 \theta \tau_{\min} \kappa_g \zeta^{-1} \sqrt{M}.$$

Proof. Let $\xi_k \in [0, 1]$ be a random variable such that $\alpha_k = \nu + \theta \xi_k \beta_k$. Then from lemma 4.5,

$$\begin{aligned}
\mathbb{E}_k[\alpha_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{\text{true}})] &= \mathbb{E}_k[(\nu + \theta\xi_k\beta_k)\tau_{\min} \nabla f(x_k)^T (d_k - d_k^{\text{true}})] \\
&= \nu \tau_{\min} \nabla f(x_k)^T \mathbb{E}[d_k - d_k^{\text{true}}] + \mathbb{E}_k[\theta \xi_k \beta_k \tau_{\min} \nabla f(x_k)^T (d_k - d_k^{\text{true}})] \\
&\leqslant \theta \tau_{\min} \mathbb{E}_k[\|\nabla f(x_k)\|_2 \|d_k - d_k^{\text{true}}\|_2] \\
&\leqslant \beta_k^2 \theta \tau_{\min} \kappa_g \zeta^{-1} \sqrt{M}.
\end{aligned}$$

■

With all the lemmas in hand, we can now prove our first main complexity result. The following theorem is stated generically for any choice of β_k.

Theorem 4.10. *Let Assumptions* 3.1, 3.3, 3.2 *hold and assume that* $\tau_{\min} > 0$ *exists. Choose* α_k *as in* (19) *and let*

$$\kappa_1 := \frac{(\nu + \theta\kappa_\beta)^2}{2}(\tau_{\min} L + \Gamma)(\zeta^{-1}\kappa_H \kappa_u^2 + \zeta^{-2} M) + \theta \kappa_g \tau_{\min} \zeta^{-1} \sqrt{M} + \theta^2 (\tau_{\min} L + \Gamma)\omega \kappa_J \kappa_c.$$

Then we have, for any $K \in \mathbb{N}$,

$$\sum_{k=0}^{K-1} \mathbb{E}[\alpha_k \beta_k \tau_{\min} (u_k^{true})^T H_k u_k^{true} + \frac{1}{2}\alpha_k \sigma \kappa_v \kappa_c^{-1} \|J_k^T c_k\|_2^2] \leqslant \tau_{\min}(f_0 - f_{\inf}) + \tau_{\min}\|c_0\|_2 + \kappa_1 \sum_{k=0}^{K-1} \beta_k^2.$$

Proof. Taking the k-th iterate conditional expectation on both sides of (33), we have

$$\begin{aligned}
&\mathbb{E}_k[\phi(x_k+\alpha_k d_k,\tau_{\min})]-\phi(x_k,\tau_{\min})\\
&\quad\leqslant \mathbb{E}_k[-\alpha_k\Delta l(x_k,\tau_{\min},\nabla f(x_k),d_k^{\text{true}})]+\frac{\alpha_k^2\beta_k^2}{2}(\tau_{\min}L+\Gamma)\mathbb{E}_k[\|u_k\|_2^2]+\frac{\alpha_k^2}{2}(\tau_{\min}L+\Gamma)\|v_k\|_2^2\\
&\quad+\mathbb{E}_k[\alpha_k\tau_{\min}\nabla f(x_k)^T(d_k-d_k^{\text{true}})]\\
&\quad\leqslant -\alpha_k\Delta l(x_k,\tau_{\min},\nabla f(x_k),d_k^{\text{true}})+\frac{\alpha_k^2\beta_k^2}{2}(\tau_{\min}L+\Gamma)(\zeta^{-1}(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\zeta^{-2}M)\\
&\quad+(\theta\kappa_g\tau_{\min}\zeta^{-1}\sqrt{M})\beta_k^2+\frac{\alpha_k^2}{2}(\tau_{\min}L+\Gamma)\|v_k\|_2^2\\
&\quad\leqslant -\alpha_k\Delta l(x_k,\tau_{\min},\nabla f(x_k),d_k^{\text{true}})+\beta_k^2[\frac{\alpha_k^2}{2}(\tau_{\min}L+\Gamma)(\zeta^{-1}(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\zeta^{-2}M)\\
&\quad+\theta\kappa_g\tau_{\min}\zeta^{-1}\sqrt{M}]+(\nu^2+\theta^2\beta_k^2)(\tau_{\min}L+\Gamma)\|v_k\|_2^2\\
&\quad\leqslant -\alpha_k\Delta l(x_k,\tau_{\min},\nabla f(x_k),d_k^{\text{true}})+\beta_k^2\kappa_1+\nu^2(\tau_{\min}L+\Gamma)\|v_k\|_2^2.
\end{aligned}$$

From (31), we have

$$\begin{aligned}
&\mathbb{E}_k[\phi(x_k+\alpha_k d_k,\tau_{\min})]-\phi(x_k,\tau_{\min})\\
&\quad\leqslant -\alpha_k(\tau_{\min}\beta_k(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\sigma(\|c_k\|_2-\|c_k+J_kd_k\|_2)+\beta_k^2\kappa_1+\nu^2(\tau_{\min}L+\Gamma)\|v_k\|_2^2\\
&\quad\leqslant -\alpha_k\tau_{\min}\beta_k(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\beta_k^2\kappa_1+(-\alpha_k\sigma+\nu^2(\tau_{\min}L+\Gamma)\frac{\omega^2\kappa_c}{\kappa_v})(\|c_k\|_2-\|c_k+J_kd_k\|_2)\\
&\quad\Big(\text{Since}\,\nu\leqslant\min\Big\{\frac{\sigma\kappa_v}{2(\tau_{\min}L+\Gamma)\omega^2\kappa_c},1-\theta\kappa_\beta\Big\}\Big)\\
&\quad\leqslant -\alpha_k\tau_{\min}\beta_k(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\beta_k^2\kappa_1-\frac{1}{2}\alpha_k\sigma(\|c_k\|_2-\|c_k+J_kd_k\|_2) \qquad (34)\\
&\quad\leqslant -\alpha_k\tau_{\min}\beta_k(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\beta_k^2\kappa_1-\frac{1}{2}\alpha_k\sigma\kappa_v\kappa_c^{-1}\|J_k^Tc_k\|_2^2.
\end{aligned}$$

Taking the total expectation of the inequality, rearranging and summing from $k=0$ to $K-1$,

$$\begin{aligned}
&\sum_{k=0}^{K-1}\mathbb{E}[\alpha_k\beta_k\tau_{\min}(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\frac{1}{2}\alpha_k\sigma\kappa_v\kappa_c^{-1}\|J_k^Tc_k\|_2^2]\\
&\qquad\leqslant\phi(x_0,\tau_{\min})-\mathbb{E}[\phi(x_K,\tau_{\min})]+\kappa_1\sum_{k=0}^{K-1}\beta_k^2.
\end{aligned}$$

Since $-\mathbb{E}[\phi(x_K,\tau_{\min})]=-\mathbb{E}[\tau_{\min}f(x_K)+\|c_K\|_2]\leqslant-\tau_{\min}f_{\text{inf}}$, we can conclude that

$$\sum_{k=0}^{K-1}\mathbb{E}[\alpha_k\beta_k\tau_{\min}(u_k^{\text{true}})^TH_ku_k^{\text{true}}+\frac{1}{2}\alpha_k\sigma\kappa_v\kappa_c^{-1}\|J_k^Tc_k\|^2]\leqslant\phi(x_0,\tau_{\min})-\tau_{\min}f_{\text{inf}}+\kappa_1\sum_{k=0}^{K-1}\beta_k^2.$$

■

This main theorem shows that, as long as we have $\tau_k\searrow\tau_{\min}>0$ and $\sum\beta_k^2\leqslant\infty$, the infeasibility measure $\|J_k^Tc_k\|_2$ converges in expectation. The convergence rate is determined

by problem-specific constants, such as L and Γ, and the starting point x_0. In fact, if the exact order of $\{\beta_k\}$ is known, we can obtain a worst-case complexity of the convergence of our infeasibility measure $\|J_k^T c_k\|_2$ and the reduced KKT measure $\|Z_k^T(\nabla f(x_k) + H_k v_k\|_2$.

Corollary 4.11. *For any $K \in \mathbb{N}$, let $\beta_k := \beta = \eta/\sqrt{K}$ for all $k \in [0, K-1]$, where $\eta > 0$. Let $\kappa_2 := \tau_{\min}(f_0 - f_{\inf}) + \|c_0\|_2 + \eta^2\kappa_1$, then*

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}[\|J_k^T c_k\|_2^2] \leqslant \frac{2\kappa_c\kappa_2}{\nu\sigma\kappa_v K}.$$

Proof. From the previous theorem, the definition of β_k and the fact that $\nu \leqslant \alpha_k$ for all $k \in \mathbb{N}$, we have

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}[\|J_k^T c_k\|_2^2] \leqslant \frac{2\kappa_c(\tau_{\min}(f_0 - f_{\inf}) + \|c_0\|_2 + \kappa_1 \sum \beta_k^2)}{\nu\sigma\kappa_v K} \leqslant \frac{2\kappa_c\kappa_2}{\nu\sigma\kappa_v K}.$$

■

Corollary 4.12. *For any $K \in \mathbb{N}$, let $\beta_k := \beta = \eta/\sqrt{K}$ for all $k \in [0, K-1]$, where $\eta > 0$. Then*

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}[\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2^2] \leqslant \frac{\kappa_H^2\kappa_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}}.$$

Proof. Similarly to Corollary 4.11, we know

$$\sum_{k=0}^{K-1} \mathbb{E}[\alpha_k\beta_k\tau_{\min}(u_k^{\text{true}})^T H_k u_k^{\text{true}}] \leqslant \phi(x_0, \tau_{\min}) - \tau_{\min} f_{\inf} + \kappa_1 \sum_{k=0}^{K-1} \beta_k^2.$$

From Lemma 4.7, we know

$$\sum_{k=0}^{K-1} \mathbb{E}[\alpha_k\beta_k\tau_{\min}\kappa_H^{-2}\zeta\|Z_k^T(\nabla f(x_k) + H_k v_k)\|^2] \leqslant \phi(x_0, \tau_{\min}) - \tau_{\min} f_{\inf} + \kappa_1 \sum_{k=0}^{K-1} \beta_k^2.$$

Substituting β_k with $\eta/\sqrt{K}$, rearranging the inequality, and dividing both sides by K, we have

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}[\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2^2] \leqslant \frac{\kappa_H^2\kappa_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}}. \tag{35}$$

■

It is obvious that for any tolerance $\epsilon_c, \epsilon_L > 0$, to achieve $\mathbb{E}\|J_k^T c_k\|_2 \leqslant \epsilon_c$, we need at most $\mathcal{O}(\epsilon_c^{-2})$ iterations. To achieve $\mathbb{E}\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2 \leqslant \epsilon_L$, we need at most $\mathcal{O}(\epsilon_L^{-4})$ iterations. To our knowledge, the former result is the best in theory when no constraint qualification is assumed over all iterations and a trust-region subproblem is utilized to solve v_k. The latter result is known to be *optimal* [17, 36] for stochastic gradient methods to find first order stationary points under our assumptions.

4.3 Case II: LICQ holds

In this subsection, we assume the following Linear Independence Constraint Qualification is satisfied for every iteration. This assumption is also called strong LICQ or everywhere LICQ. The analysis is parallel to Section 4.2, while certain improvements are made to the convergence of the infeasibility measure, and the reduced gradient measure is replaced with the KKT measure. Most of the existing literature assumes LICQ or other types of constraint qualification in their analysis, such as the mean-squared smoothness constraint qualification in [31] and the MFCQ in [10].

Assumption 4.13. *(LICQ) For all $k \in \mathbb{N}$, the constraint Jacobian matrix $J(x_k)$ has full row rank. Equivalently, there exists a positive constant $\sigma_J > 0$ such that $\|J_k^T c_k\|_2 \geqslant \sigma_J \|c_k\|_2$.*

The following lemma is an improved version of Lemma 4.2. Since LICQ holds for all iterations, we can now use the exact infeasibility measure $\varphi = \|c(x)\|_2$.

Lemma 4.14. *Let assumption* 4.13 *hold for all $k \in \mathbb{N}$, then with $\|c_k\|_2 > 0$,*

$$\|c_k\|_2 - \|c_k + J_k v_k\|_2 \geqslant \kappa_v \sigma_J^2 \|c_k\|_2. \tag{36}$$

Proof. From Lemma 4.2,

$$\|c_k\|_2(\|c_k\|_2 - \|c_k + J_k v_k\|_2) \geqslant \kappa_v \|J_k^T c_k\|_2^2 \geqslant \kappa_v \sigma_J^2 \|c_k\|_2^2.$$

Dividing both sides by $\|c_k\|_2$ yields the result. ■

The following Lemma shows that LICQ serves as a sufficient condition for the existence of $\tau_{\min} > 0$, which makes our analysis in the previous subsection meaningful.

Lemma 4.15. *Let $\sigma \in (0,1)$ and $\beta_k \leqslant \kappa_\beta$ hold for all $k \in \mathbb{N}$. Let Assumptions* 3.1, 3.3, 3.2 *and* 4.13 *hold. Then, there exists such $\tau_{\min} := \dfrac{(1-\sigma)\kappa_v \sigma_J}{\omega(\kappa_\beta \kappa_H \kappa_u + \kappa_g)}$ satisfying*

$$\tau_{\min}(\nabla f(x_k)^T d_k^{true} + \beta_k (u_k^{true})^T H_k u_k^{true}) \leqslant (1-\sigma)(\|c_k\|_2 - \|c_k + J_k d_k\|_2) \text{ for all } k \in \mathbb{N}.$$

Proof. By the definition of d_k^{true},

$$\begin{aligned}\nabla f(x_k)^T d_k^{\text{true}} &= \nabla f(x_k)^T (\beta_k u_k^{\text{true}} + v_k) \\ &= -\beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} - \beta_k v_k^T H_k u_k^{\text{true}} + \nabla f(x_k)^T v_k.\end{aligned}$$

By the trust-region radius of subproblem (15), we have

$$\begin{aligned}\nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} &= -\beta_k v_k^T H_k u_k^{\text{true}} + \nabla f(x_k)^T v_k \\ &\leqslant (\beta_k \kappa_H \|u_k^{\text{true}}\|_2 + \|\nabla f(x_k)\|_2)\|v_k\|_2 \\ &\leqslant (\kappa_\beta \kappa_H \kappa_u + \kappa_g)\omega \|J_k^T c_k\|_2 \\ &\leqslant \omega(\kappa_\beta \kappa_H \kappa_u + \kappa_g)\frac{1}{\sigma_J \kappa_c}\frac{\kappa_c}{\kappa_v}(\|c_k\|_2 - \|c_k + J_k d_k\|_2) \\ &= \omega(\kappa_\beta \kappa_H \kappa_u + \kappa_g)\frac{(\|c_k\|_2 - \|c_k + J_k d_k\|_2)}{\kappa_v \sigma_J}.\end{aligned}$$

Then multiplying $\tau_{\min}$ on both sides proves the result. ■

Lemma 4.16. *Let Assumptions* 3.1*,* 3.2*, and* 4.13 *hold. Then we have*

$$(u_k^{true})^T H_k(u_k^{true}) \geqslant \zeta\kappa_H^{-2}\|\nabla f(x_k) + J_k^T y_k^{true}\|_2^2 - \kappa_3(\|c_k\|_2 - \|c_k + J_k d_k\|_2),$$

where $\kappa_3 := \zeta(2\kappa_u + \omega\kappa_J\kappa_c)\dfrac{\omega}{\kappa_v\sigma_J}$.

Proof. By Assumption 3.2,

$$(u_k^{\text{true}})^T H_k(u_k^{\text{true}}) \geqslant \zeta\|u_k^{\text{true}}\|_2^2 \geqslant \zeta\kappa_H^{-2}\|H_k u_k^{\text{true}}\|_2^2.$$

Then, by (21) and Lemma 4.3, we have

$$\begin{aligned}
\|H_k u_k^{\text{true}}\|_2^2 &= \|H_k u_k^{\text{true}} + H_k v_k - H_k v_k\|_2^2 \\
&= \|H_k u_k^{\text{true}} + H_k v_k\|_2^2 - 2v_k^T H_k H_k(u_k^{\text{true}} + v_k) + \|H_k v_k\|_2^2 \\
&\geqslant \|\nabla f(x_k) + J_k^T y_k^{\text{true}}\|_2^2 - 2v_k^T H_k H_k u_k^{\text{true}} - \|H_k v_k\|_2^2 \\
&\geqslant \|\nabla f(x_k) + J_k^T y_k^{\text{true}}\|_2^2 - \kappa_H^2(2\|u_k^{\text{true}}\|_2 + \|v_k\|_2)\|v_k\|_2 \\
&\geqslant \|\nabla f(x_k) + J_k^T y_k^{\text{true}}\|_2^2 - \kappa_H^2(2\kappa_u + \omega\kappa_J\kappa_c)\frac{\omega}{\kappa_v\sigma_J}(\|c_k\|_2 - \|c_k + J_k d_k\|_2) \\
&\geqslant \|\nabla f(x_k) + J_k^T y_k^{\text{true}}\|_2^2 - \kappa_H^2\zeta^{-1}\kappa_3(\|c_k\|_2 - \|c_k + J_k d_k\|_2).
\end{aligned}$$

■

Lemma 4.16 is parallel to Lemma 4.7, while the reduced gradient $\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2$ is replaced by the true gradient of the Lagrangian function. With these lemmas, we can further obtain a primal-dual convergence result.

Theorem 4.17. *Let Assumptions* 3.1*,* 3.3*,* 3.2 *and* 4.13 *hold for all* $k \in \mathbb{N}$*. Choose* $\beta_k = \eta/\sqrt{K}$ *and* α_k *as in* (19)*. Then for any* $K \in \mathbb{N}$*,*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|c_k\|_2 \leqslant \frac{2\kappa_2}{\nu\sigma\kappa_v\sigma_J^2 K}, \quad \textit{and}$$

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla f(x_k) + J_k^T y_k^{true}\|_2^2] \leqslant \frac{\kappa_H^2\kappa_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}} + \frac{2\kappa_H^2\kappa_2\kappa_3}{\nu\sigma\kappa_v\sigma_J^2\zeta K}.$$

Proof. By Lemma 4.15, we know that $\tau_{\min} > 0$ exists. Therefore, the conditions for theorem 4.10 are perfectly satisfied, and we have the following result from (34).

$$\sum_{k=0}^{K-1}\mathbb{E}[\alpha_k\beta_k\tau_{\min}(u_k^{\text{true}})^T H_k u_k^{\text{true}} + \frac{1}{2}\alpha_k\sigma(\|c_k\|_2 - \|c_k + J_k d_k\|_2)] \leqslant \phi(x_0, \tau_{\min}) - \tau_{\min} f_{\inf} + \kappa_1\eta^2.$$

From Lemma 4.14, we have

$$\sum_{k=0}^{K-1}\frac{1}{2}\nu\sigma\kappa_v\sigma_J^2\|c_k\|_2 \leqslant \kappa_2.$$

Divide both sides by K, and we get the first result

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|c_k\|_2 \leqslant \frac{2\kappa_2}{\nu\sigma\kappa_v\sigma_J^2 K}. \tag{37}$$

From lemma 4.16, we know

$$\sum_{k=0}^{K-1}\mathbb{E}[\nu\beta_k\tau_{\min}\zeta\kappa_H^{-2}\|\nabla f(x_k)+J_k^T y_k^{\text{true}}\|_2^2+(\frac{1}{2}\nu\sigma-\nu\beta_k\tau_{\min}\kappa_3)(\|c_k\|_2-\|c_k+J_k d_k\|_2)]$$
$$\leqslant \sum_{k=0}^{K-1}\mathbb{E}[\alpha_k\beta_k\tau_{\min}(u_k^{\text{true}})^T H_k u_k^{\text{true}}+\frac{1}{2}\alpha_k\sigma(\|c_k\|_2-\|c_k+J_k d_k\|_2)]\leqslant\kappa_2.$$

Therefore,

$$\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla f(x_k)+J_k^T y_k^{\text{true}}\|_2^2]\leqslant\frac{\kappa_H^2\kappa_2\sqrt{K}}{\nu\eta\tau_{\min}\zeta}+\frac{\kappa_3\kappa_H^2}{\zeta}\sum_{k=0}^{K-1}\mathbb{E}\|c_k\|_2. \tag{38}$$

Dividing both sides by K and by (37) can yield the result

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla f(x_k)+J_k^T y_k^{\text{true}}\|_2^2]\leqslant\frac{\kappa_H^2\kappa_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}}+\frac{2\kappa_H^2\kappa_2\kappa_3}{\nu\sigma\kappa_v\sigma_J^2\zeta K}.$$

■

Parallel to Corollary 4.11 and 4.12, we can obtain worst-case complexity results from Theorem 4.10. To achieve $\mathbb{E}\|c_k\|_2\leqslant\epsilon_c$, we only need at most $\mathcal{O}(\epsilon_c^{-1})$ iterations. This result is the same as the complexity of the infeasibility measure in [28] (although it uses a l_1-norm and an exact u_k), which is optimal under a constraint qualification, to our knowledge. Apparently, the improvement is attributed to Assumption 4.13.

To achieve $\mathbb{E}\|\nabla f(x_k)+J_k^T y_k^{\text{true}}\|_2\leqslant\epsilon_L$, at most $\mathcal{O}(\epsilon_L^{-4})$ iterations are still needed. However, we can now use the true KKT measure $\|\nabla f(x_k)+J_k^T y_k^{\text{true}}\|_2$, which improves the reduced gradient measure in Corollary 4.12.

4.4 Case III: merit parameter sequence vanishes

For the analysis in the previous subsections, we assumed that $\tau_{\min}>0$ exists, even though our algorithm does not directly compute it. Unfortunately, in certain cases, such $\tau_{\min}$ does not exist, *i.e.,*the merit parameter sequence $\{\tau_k\}$ may decrease to 0 in the limit. The only reason for this kind of behavior is that a (sub)sequence of constraint Jacobians tends toward rank deficiency, where the best we can hope for is the convergence of $\|J_k^T c_k\|_2$.

Since we do not have $\tau_{\min}$ now, we need to make some changes to the parameters. As α_k remains in $[\nu,\nu+\theta\beta_k]$, ν is now selected from a different range:

$$\nu\in(0,\min\{\frac{\sigma\kappa_v\kappa_c^{-1}}{2\Gamma\omega^2},1-\theta\kappa_\beta\}].$$

Lemma 4.18. *Under Assumptions* 3.1, 3.3, *and* 3.2, *it follows for all* $k \in \mathbb{N}$ *that*

$$\mathbb{E}_k[\|c_k\|_2 - \|c(x_k + \alpha_k d_k)\|_2] \geqslant \frac{1}{2}\nu\sigma\kappa_v\kappa_c^{-1}\mathbb{E}_k\|J_k^T c_k\|_2^2 - \gamma_1\tau_k - \gamma_2\beta_k^2 - \gamma_3\beta_k^2\tau_k \tag{39}$$

where we define

$$\gamma_1 := f_{\sup} - f_{\inf} + \frac{\sigma^2\kappa_v^2\kappa_J^2 L}{4\Gamma^2\omega^2},$$

$$\gamma_2 := \frac{1}{2}\Gamma\kappa_u^2 + \Gamma\theta^2\kappa_J^2\kappa_c^2,$$

$$\gamma_3 := \frac{1}{2}L\kappa_u^2 + L\theta^2\kappa_J^2\kappa_c^2 + \theta\kappa_g\zeta^{-1}\sqrt{M}.$$

Proof. From the definition of $\phi(x,\tau)$, we have

$$\phi(x_k,\tau_k) - \phi(x_k + \alpha_k d_k) = \tau_k(f(x_k) - f(x_k + \alpha_k d_k)) + (\|c_k\|_2 - \|c(x_k + \alpha_k d_k)\|_2)$$

Hence,

$$\begin{aligned}
&\|c_k\|_2 - \|c(x_k + \alpha_k d_k)\|_2 = \phi(x_k,\tau_k) - \phi(x_k + \alpha_k d_k) - \tau_k(f(x_k) - f(x_k + \alpha_k d_k)) \\
&\geqslant -\alpha_k\tau_k\nabla f(x_k)^T d_k + \alpha_k(\|c_k\|_2 - \|c_k + J_k d_k\|_2) - \frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2\|d_k\|_2^2 - \tau_k[f(x_k) - f(x_k + \alpha_x d_k)] \\
&= \underbrace{-\alpha_k\tau_k\nabla f(x_k)^T d_k^{\text{true}} + \alpha_k(\|c_k\|_2 - \|c_k + J_k d_k\|_2)}_{(I)} \underbrace{-\frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2(\beta_k^2\|u_k\|_2^2 + \|v_k\|_2^2)}_{(II)} \\
&\qquad \underbrace{-\tau_k[f(x_k) - f(x_k + \alpha_x d_k)]}_{(III)} \underbrace{-\alpha_k\tau_k\nabla f(x_k)^T(d_k - d_k^{\text{true}})}_{(IV)}
\end{aligned}$$

Now we break this into four parts:

$$\begin{aligned}
(I) &= \alpha_k\Delta l(x_k,\tau_k,\nabla f(x_k), d_k^{\text{true}}) \\
&\geqslant \alpha_k\tau_k(u_k^{\text{true}})^T H_k u_k^{\text{true}} + \alpha_k\sigma(\|c_k\|_2 - \|c_k + J_k d_k\|_2) \geqslant \nu\sigma\kappa_v\kappa_c^{-1}\|J_k^T c_k\|_2^2.
\end{aligned}$$

$$\begin{aligned}
(II) = -\frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2(\beta_k^2\|u_k\|_2^2 + \|v_k\|_2^2) &= -\frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2\kappa_u^2\beta_k^2 - \frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2\|v_k\|_2^2 \\
&\geqslant -\frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2\kappa_u^2\beta_k^2 - \frac{1}{2}(\tau_k L + \Gamma)\alpha_k^2\omega^2\|J_k^T c_k\|_2^2 \\
&\geqslant -\frac{1}{2}(\tau_k L + \Gamma)\kappa_u^2\beta_k^2 - \frac{1}{2}(\tau_k L + \Gamma)(\nu + \theta\beta_k)^2\omega^2\|J_k^T c_k\|_2^2 \\
&\geqslant -\frac{1}{2}(\tau_k L + \Gamma)\kappa_u^2\beta_k^2 - (\tau_k L + \Gamma)(\nu^2 + \theta^2\beta_k^2)\omega^2\|J_k^T c_k\|_2^2 \\
&\geqslant -\frac{1}{2}(\tau_k L + \Gamma)\kappa_u^2\beta_k^2 - (\tau_k L + \Gamma)\nu^2\omega^2\|J_k^T c_k\|_2^2 - (\tau_k L + \Gamma)\theta^2\kappa_J^2\kappa_c^2\beta_k^2.
\end{aligned}$$

$$\mathbb{E}_k[(IV)] \geqslant -\beta_k^2\theta\tau_k\kappa_g\zeta^{-1}\sqrt{M}.$$

Adding all four parts together and take expectation, we have

$$\begin{aligned}\mathbb{E}_k[\|c_k\|_2 - \|c(x_k+\alpha_k d_k)\|_2] &\geqslant \nu\sigma\kappa_v\kappa_c^{-1}\mathbb{E}_k\|J_k^T c_k\|_2^2 - (\tau_k L+\Gamma)\nu^2\omega^2\mathbb{E}_k\|J_k^T c_k\|_2^2 - \tau_k(f_k - f_{\inf}) \\ &\quad - \beta_k^2\Big[\frac{1}{2}(\tau_k L+\Gamma)\kappa_u^2 + (\tau_k L+\Gamma)\theta^2\kappa_J^2\kappa_c^2 + \theta\tau_k\kappa_g\zeta^{-1}\sqrt{M}\Big] \\ &\geqslant \frac{1}{2}\nu\sigma\kappa_v\kappa_c^{-1}\mathbb{E}_k\|J_k^T c_k\|_2^2 - \tau_k(f_{\sup} - f_{inf} + L\nu^2\omega^2\kappa_J^2\kappa_c^2) \\ &\quad - \beta_k^2(\frac{1}{2}\Gamma\kappa_u^2 + \Gamma\theta^2\kappa_J^2\kappa_c^2) - \beta_k^2\tau_k(\frac{1}{2}L\kappa_u^2 + L\theta^2\kappa_J^2\kappa_c^2 + \theta\kappa_g\zeta^{-1}\sqrt{M}) \\ &= \frac{1}{2}\nu\sigma\kappa_v\kappa_c^{-1}\mathbb{E}_k\|J_k^T c_k\|_2^2 - \gamma_1\tau_k - \gamma_2\beta_k^2 - \gamma_3\beta_k^2\tau_k.\end{aligned}$$

■

Theorem 4.19. *Let Assumptions* 3.1, 3.3, *and* 3.2 *hold for all* $k \in \mathbb{N}$, *and assume the merit parameter sequence* $\{\tau_k\} \searrow 0$. *If* $\{\beta_k\}$ *is chosen so that* $\Sigma_{k=1}^{\infty}\beta_k^2 < \infty$, $\liminf_{k\to\infty} \|J_k^T c_k\|_2 = 0$.

Proof. Since $\tau_k \searrow 0$, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\tau_k < \varepsilon/\gamma_1$ and $\tau_k < \varepsilon/\gamma_3$ for all $k > N$. Summing (39) from N to ∞, taking total expectation and telescoping, we have

$$\begin{aligned}\frac{1}{2}\nu\sigma\kappa_v\kappa_c^{-1}\sum_{k=N}^{\infty}\mathbb{E}\|J_k^T c_k\|_2^2 &\leqslant \|c_N\|_2 + \sum_{k=N}^{\infty}(\gamma_1\tau_k + \gamma_2\beta_k^2 + \gamma_3\beta_k^2\tau_k) \\ &\leqslant \kappa_c + \sum_{k=N}^{\infty}\varepsilon + (\varepsilon+\gamma_2)\sum_{k=N}^{\infty}\beta_k^2.\end{aligned} \tag{40}$$

Taking $\varepsilon \to 0$, we have $\liminf_{k\to\infty} \mathbb{E}_k\|J_k^T c_k\|_2 = 0$. ■

Unlike the first two cases, we cannot obtain a worst-case complexity result without knowing the actual decay rate of τ_k. The lim inf type of convergence is common in stochastic optimization literature, as Theorem 3 in [2].

5 An Inexact Stochastic SQP Algorithm

In our previous discussion, we assumed that an exact solution of the subproblem (17) is achieved so that condition $u_k \in \text{Null}(J_k)$, *i.e.,*$J_k u_k = 0$ is strictly satisfied. However, such an exact u_k is expensive to compute in practice. More importantly, since u_k is solved with the stochastic subproblem 17 involving g_k, it is natural that we should not put too much effort on solving u_k to optimum, since it would be *optimal with error*. Therefore, we turn to inexact solvers for 17. In our analysis, termination tests (TT) are used to track the residuals as a stopping criterion for iterative linear system solvers such as MINRES. Accordingly, the algorithm is reformulated as algorithm 4.

In this section, we will first define our termination tests to solve the subproblem 17. Then, we prove convergence and complexity results when an inexact solver is utilized. Now we introduce and (re)define some variables for the analysis in this section.

From now on, u_k and y_k are solved via an inexact iterative linear system solver, where the solver yields a pair of residuals $(\rho_k, r_k) \in \mathbb{R}^n \times \mathbb{R}^m$ such that for every iteration $k \in \mathbb{N}$, (ρ_k, r_k) satisfies our termination tests. $u_{k,*}$ is defined as the exact solution of the subproblem 17 and lies in the Null space of J_k (the u_k we use in the sections above). Recall the general KKT system of subproblem (18), we define the (primal-dual) residual pair (ρ_k, r_k) formally as follows:

$$\begin{bmatrix} \rho_k \\ r_k \end{bmatrix} := \begin{bmatrix} H_k & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} u_k \\ y_k \end{bmatrix} + \begin{bmatrix} g_k + H_k v_k \\ 0 \end{bmatrix}. \tag{41}$$

We make the following assumptions on the iterative solver we use. Note that this is a common property of linear system solvers. In fact, Krylov Subspace solvers, such as MINRES, can produce an exact solution after $t = n$ iterations.

Assumption 5.1. *For all $k \in \mathbb{N}$ in any run, the iterative system solver used to compute u_k and y_k generates a sequence $\{(u_{k,t}, \rho_{k,t}, r_{k,t})\}_{t \geqslant 0}$ with*

$$\begin{bmatrix} \rho_{k,t} \\ r_{k,t} \end{bmatrix} := \begin{bmatrix} H_k & J_k^T \\ J_k & 0 \end{bmatrix} \begin{bmatrix} u_{k,t} \\ y_{k,t} \end{bmatrix} + \begin{bmatrix} g_k + H_k v_k \\ 0 \end{bmatrix}, \tag{42}$$

such that $\lim_{t \to \infty} \|(u_{k,t}, \rho_{k,t}, r_{k,t}) - (u_{k,*}, 0, 0)\| = 0$.

Algorithm 4 A Two Stepsize Inexact Stochastic SQP Algorithm

Initialize $x_0 \in \mathbb{R}^n$.
for $k = 1, 2, 3, ..., K$ **do**
  Compute normal component v_k with (15).
  Compute stochastic gradient g_k.
  Choose $\beta_k > 0$.
  Run the iterative solver until the Termination Test is satisfied to get u_k.
  Set $d_k \leftarrow \beta_k u_k + v_k$.
  Choose $\alpha_k > 0$.
  Set $x_{k+1} \leftarrow x_k + \alpha_k d_k$.
end for

Formally, we ask for the following termination test of the iterative linear system solver.

Termination Test 1. *Given positive constants $(\gamma_r, \gamma_\rho) \in \mathbb{R}_{>0} \times \mathbb{R}_{>0}$. β_k is the predefined positive rescaling parameter. For any $k \in \mathbb{N}$ and any run of the algorithm, we terminate the iterative linear solver as long as the residuals satisfy the following conditions:*

$$\|r_k\|_2 \leqslant \gamma_r \beta_k \quad \textit{and} \quad \|\rho_k\|_2 \leqslant \gamma_\rho \beta_k.$$

Under Assumption 5.1, the termination test is bound to be satisfied since (r_k, ρ_k) converges to 0 as $t \longrightarrow \infty$.

Since our inexact u_k does not fall into the Null space of J_k now, we need some parallel lemmas to Lemma 4.6 to continue our convergence analysis. First, we define the Moore-Penrose inverse of J_k with the SVD form. Assume $\text{rank}(J_k) = r$, and its singular values are denoted as $(\sigma_1, \sigma_2, \ldots, \sigma_r, 0, \ldots, 0)$, where $\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_r$. Then we have the singular value decomposition of J_k as $J_k = U_k \Sigma_k V_k^T$, where $\Sigma_k \in \mathbb{R}^{m \times n}$. $U_k \in \mathbb{R}^{m \times m}$ and $V_k \in \mathbb{R}^{n \times n}$ are orthogonal matrices. When $r \leqslant m$, we also have a compact SVD form, $J_k = U_{k,r} \Sigma_{k,r} V_{k,r}^T$,

where $\Sigma_{k,r} = \text{diag}(\sigma_1, \dots \sigma_r)$ keeps only positive singular values. $U_{k,r} \in \mathbb{R}^{m\times r}$ and $V_{k,r} \in \mathbb{R}^{n\times r}$ are the first r columns of U_k and V_k respectively. Then we can define the pseudo-inverse of J_k as $J_k^+ = V_{k,r}\Sigma_{k,r}^{-1}U_{k,r}^T$.

Remark. Since we do not assume the full row rank of J_k, we can only use a pseudo-inverse *i.e.,*J_k^+ here. For the matrix $(J_kJ_k^T)$ is not guaranteed to be invertible, J_k^+ is a natural generalization of $J_k^T(J_kJ_k^T)^{-1}$. When J_k is invertible, the two forms are the same.

We now derive the explicit forms of inexact u_k, exact $u_{k,*}$, and true u_k^{true}. With J_k^+, we can write u_k as the decomposition $u_k = u_{k,1} + u_{k,2}$, where $u_{k,1} \in \text{Range}(J_k)$ and $u_{k,2} \in \text{Null}(J_k)$. From (41), we have

$$u_{k,1} = J_k^+ r_k, \quad u_{k,2} = -Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k + H_k v_k + H_k J_k^+ r_k - \rho_k) \tag{43}$$

The *inexact* solution of the subproblem we are now using is

$$u_k = J_k^+ r_k - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k + H_k v_k + H_k J_k^+ r_k - \rho_k).$$

The *exact* solution of the stochastic subproblem is

$$u_{k,*} = -Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k + H_k v_k).$$

The tangential component computed with the *true* gradient is

$$u_k^{\text{true}} = -Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (\nabla f(x_k) + H_k v_k).$$

To derive a uniform bound on $\|u_k\|_2^2$, we make the following assumption on the behavior of constraint Jacobian matrices J_k's.

Assumption 5.2. *For all $k \in \mathbb{N}$, the nonzero singular values of J_k is uniformly bounded by some positive value $\bar{\sigma}_{\min} \in \mathbb{R}_{>0}$.*

This assumption covers the LICQ case, where all the singular values of the constraint Jacobians are positive and bounded from below. To clarify, since the assumption does not require all singular values to be positive, it includes certain "good" cases of rank-deficient Jacobians. This assumption is made primarily due to analysis requirements. Practically, one can simply set a small error tolerance to ensure that the assumption is satisfied. Assumption 5.2 yields a uniform upper bound that is $\|J_k^+\| \leqslant 1/\bar{\sigma}_{\min}$ for all $k \in \mathbb{N}$. With Assumption 5.2, we have the following lemma.

Lemma 5.3. *For all $k \in \mathbb{N}$, $\|\mathbb{E}_k[u_k - u_k^{true}]\|_2 \leqslant \kappa_4\beta_k$, where $\kappa_4 := (\frac{1}{\bar{\sigma}_{\min}}\gamma_r + \zeta^{-1}\gamma_\rho)$.*

Proof.

$$\begin{aligned}\mathbb{E}_k[u_k - u_k^{\text{true}}] &= \mathbb{E}_k[J_k^+ r_k - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T (g_k - \nabla f(x_k)) \\ &\quad - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T H_k J_k^+ r_k + Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T \rho_k] \\ &= \mathbb{E}_k[(I - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T H_k) J_k^+ r_k] + \mathbb{E}_k[Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T \rho_k].\end{aligned}$$

From the proof of Lemma 11 in [2], we know that $\|I - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T H_k\|_2 \leqslant 1$ and $\|Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T\|_2 \leqslant \zeta^{-1}$. Taking expectations on both sides and using the triangle inequality, we have

$$\begin{aligned}\|\mathbb{E}_k(u_k - u_k^{\text{true}})\|_2 &\leqslant \|\mathbb{E}_k[(I - Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T H_k) J_k^+ r_k]\|_2 + \|\mathbb{E}_k[Z_k(Z_k^T H_k Z_k)^{-1} Z_k^T \rho_k]\|_2 \\ &\leqslant \frac{1}{\bar{\sigma}_{\min}}\gamma_r\beta_k + \zeta^{-1}\gamma_\rho\beta_k = \kappa_4\beta_k.\end{aligned}$$

■

Lemma 5.4. *For all $k \in \mathbb{N}$, $\mathbb{E}_k\|u_k\|_2^2$ is bounded from above by $\tilde{\kappa}_u := 2\kappa_u^2 + 2(\zeta^{-1}\sqrt{M} + \kappa_4\beta_k)^2$.*

Proof. From Lemma 11 in [2], we know that $\mathbb{E}_k[\|u_k - u_k^{\text{true}}\|] \leqslant \zeta^{-1}\sqrt{M} + \kappa_4\beta_k$. Hence,

$$\mathbb{E}_k\|u_k\|_2^2 \leqslant \mathbb{E}_k[2(\|u_k^{\text{true}}\|_2^2 + \|u_k - u_k^{\text{true}}\|_2^2)] \leqslant 2\kappa_u^2 + 2(\zeta^{-1}\sqrt{M} + \kappa_4\beta_k)^2 = \tilde{\kappa}_u.$$

■

We still use the merit parameter updating scheme (24), while the constraint violation reduction is modified to $\|c_k\|_2 - \|c_k + J_k v_k\|_2$. We restate it here: for some fixed $\sigma \in (0,1)$ and $\varepsilon_\tau \in [0,1)$, let

$$\tau_k^{trial} \leftarrow \begin{cases} \infty & \text{if } \nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} \leqslant 0 \\ \dfrac{(1-\sigma)(\|c_k\|_2 - \|c_k + J_k v_k\|_2)}{\nabla f(x_k)^T d_k^{\text{true}} + \beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}}} & \text{otherwise,} \end{cases} \tag{44}$$

and

$$\tau_k \leftarrow \begin{cases} \tau_{k-1} & \text{if } \tau_{k-1} \leqslant \tau_k^{\text{trial}} \\ \min\{(1-\varepsilon_\tau)\tau_{k-1}, \tau_k^{\text{trial}}\} & \text{otherwise.} \end{cases}$$

Since $\Delta l(x_k, \tau_k, \nabla f(x_k), d_k^{\text{true}}) = -\tau_k \nabla f_k^T d_k^{\text{true}} + (\|c_k\|_2 - \|c_k + J_k v_k\|_2)$, if there exists $\tau_{\min} > 0$, we have

$$\Delta l(x_k, \tau_{\min}, \nabla f(x_k), d_k^{\text{true}}) \geqslant \tau_{\min}\beta_k (u_k^{\text{true}})^T H_k u_k^{\text{true}} + \sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2),$$

which is analogous to (26).

Theorem 5.5. *Let Assumptions* 3.1*,* 3.3*,* 3.2*,* 5.1*, and* 5.2 *hold. Use an inexact solver for the problem* 17*. Choose $\beta_k = \eta/\sqrt{K}$ and $\alpha_k \in [\nu, \nu + \theta\beta_k]$. Define $\tilde{\kappa}_1 := \tau_{\min}\kappa_g\kappa_4 + (\tau_{\min}L + \Gamma)\tilde{\kappa}_u + \theta^2(\tau_{\min}L + \Gamma)\omega^2\kappa_J^2\kappa_c^2 + \gamma_r$ and $\tilde{\kappa}_2 = \tau_{\min}(f_{\sup} - f_{\inf}) + \|c_0\|_2 + \eta^2\tilde{\kappa}_1$, then*

$$\sum_{k=0}^{K-1} \mathbb{E}[\nu\beta_k\tau_{\min}(u_k^{true})^T H_k u_k^{true} + \frac{1}{2}\nu\sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2)] \leqslant \tilde{\kappa}_2. \tag{45}$$

Proof. Consider the reduction in merit function ϕ, we have

$$\begin{aligned}\phi(x_k + \alpha_k d_k, \tau_{\min}) - \phi(x_k, \tau_{\min}) \leqslant &-\alpha_k \Delta l(x_k, \tau_{\min}, \nabla f_k, d_k^{\text{true}}) - \alpha_k\tau_{\min}\nabla f_k^T(d_k^{\text{true}} - d_k) \\ &-\alpha_k(\|c_k + J_k v_k\|_2 - \|c_k + J_k v_k + \beta_k r_k\|_2 + \frac{1}{2}(\tau_{\min}L + \Gamma)\nabla f_k^T(d_k^{\text{true}} - d_k).\end{aligned}$$

Using $\|d_k\|_2^2 \leqslant 2(\beta_k^2\|u_k\|_2^2 + \|v_k\|_2^2)$ and rearranging the order of the inequality, we have

$$\begin{aligned}\alpha_k \Delta l(x_k, \tau_{\min}\nabla f_k, d_k^{\text{true}}) \leqslant &(\phi(x_k, \tau_{\min}) - \phi(x_k + \alpha_k d_k, \tau_{\min})) + \alpha_k\tau_{\min}\nabla f_k^T(d_k - d_k^{\text{true}}) \\ &+\alpha_k(\|c_k + J_k v_k + \beta_k r_k\|_2 - \|c_k + J_k v_k\|_2) + (\tau_{\min}L + \Gamma)\alpha_k^2(\beta_k^2\|u_k\|_2^2 + \|v_k\|_2^2).\end{aligned}$$

Since $\nu \in (0, \min\{\frac{\sigma\kappa_v\kappa_c^{-1}}{4(\tau_{\min}L+\Gamma)\omega^2}, 1-\theta\kappa_\beta\}]$ and $\alpha_k \in [\nu, \nu+\theta\beta_k]$, taking the expectation, we have

$$\begin{aligned}\mathbb{E}_k\nu[\tau_{\min}\beta_k(u_k^{\text{true}})^T H_k u_k^{\text{true}} + \sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2)] \leqslant \mathbb{E}_k[(\phi(x_k, \tau_{\min}) - \phi(x_k + \alpha_k d_k, \tau_{\min})] \\ + \mathbb{E}_k[\alpha_k \tau_{\min} \nabla f_k^T (d_k - d_k^{\text{true}})] + \mathbb{E}_k[\alpha_k(\|c_k + J_k v_k\|_2 - \|c_k + J_k v_k\|_2 + \beta_k\|r_k\|_2)] \\ + \mathbb{E}_k[\alpha_k^2\beta_k^2(\tau_{\min}L+\Gamma)\|u_k\|_2^2] + \nu^2(\tau_{\min}L+\Gamma)\omega^2\|J_k^T c_k\|_2^2 + \theta^2\beta_k^2(\tau_{\min}L+\Gamma)\omega^2\kappa_J^2\kappa_c^2.\end{aligned}$$

Since $\alpha_k \leqslant 1$ and $\|r_k\| \leqslant \gamma_r\beta_k$, we know

$$\mathbb{E}_k\alpha_k\beta_k\|r_k\| \leqslant \gamma_r\beta_k^2.$$

By the choice of ν and (4.2) we have

$$\nu^2(\tau_{\min}L+\Gamma)\omega^2\|J_k^T c_k\|_2^2 \leqslant \frac{1}{2}\nu\sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2).$$

Combining the results in Lemma 5.3 and 5.4, we can conclude that

$$\begin{aligned}\mathbb{E}_k[\nu\tau_{\min}\beta_k(u_k^{\text{true}})^T H_k u_k^{\text{true}} + \frac{1}{2}\nu\sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2)] \leqslant \mathbb{E}_k[\phi(x_k, \tau_{\min}) - \phi(x_k + \alpha_k d_k, \tau_{\min})] \\ + \beta_k^2(\tau_{\min}\kappa_g(\frac{1}{\bar{\sigma}_{\min}}\gamma_r + \zeta^{-1}\gamma_\rho) + (\tau_{\min}L+\Gamma)\tilde{\kappa}_u + \theta^2(\tau_{\min}L+\Gamma)\omega^2\kappa_J^2\kappa_c^2 + \gamma_r) \\ = \mathbb{E}_k[\phi(x_k, \tau_{\min}) - \phi(x_k + \alpha_k d_k, \tau_{\min})] + \tilde{\kappa}_1\beta_k^2.\end{aligned}$$

Similarly, summing from $k = 0$ to $K-1$, and substituting $\beta_k = \eta/\sqrt{K}$, we have

$$\sum_{k=0}^{K-1} \mathbb{E}[\nu\beta_k\tau_{\min}(u_k^{\text{true}})^T H_k u_k^{\text{true}} + \frac{1}{2}\nu\sigma(\|c_k\|_2 - \|c_k + J_k v_k\|_2)] \leqslant \tilde{\kappa}_2.$$

■

Theorem 5.5 proves results in parallel to Theorem 4.10. The difference lies mainly in the constants. $\tilde{\kappa}_1$ includes additional terms that include γ_ρ and γ_r. This is natural in that our Termination Test 1 forces the errors of residuals below a threshold that is controlled by β_k. Unfortunately, this β_k is inevitable due to the need for higher accuracy as the iterations approach stationarity. If we remove β_k in the Termination Test 1, the algorithm can only guarantee convergence to a neighborhood of the optimal solution or a stationary point.

For the sake of brevity, we only list the complexity results of the inexact algorithm under two different cases. Their proofs follow from Corollary4.11, 4.12, and Theorem 4.17. As for the rank-deficient case, a similar lim inf convergence result to Theorem 4.19 can be obtained.

Corollary 5.6. *Assume there exists $\tau_{\min} \in \mathbb{R}_{>0}$. Under the same assumptions as in Theorem 5.5, we have the following results:*

I *When no constraint qualification is assumed, we have*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|J_k^T c_k\|_2^2] \leqslant \frac{2\kappa_c\tilde{\kappa}_2}{\nu\sigma\kappa_v K} \quad \textit{and} \quad \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|Z_k^T(\nabla f(x_k) + H_k v_k)\|_2^2] \leqslant \frac{\kappa_H^2\tilde{\kappa}_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}}.$$

II *When LICQ holds for all iterate k, we have*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|c_k\|_2] \leqslant \frac{2\tilde{\kappa}_2}{\nu\sigma\kappa_v\sigma_J^2 K}, \text{ and}$$
$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|\nabla f(x_k) + J_k^T y_k^{true}\|_2^2] \leqslant \frac{\kappa_H^2\tilde{\kappa}_2}{\nu\eta\tau_{\min}\zeta\sqrt{K}} + \frac{2\kappa_H^2\tilde{\kappa}_2\kappa_3}{\nu\sigma\kappa_v\sigma_J^2\zeta K}.$$

Again, we need at most $\mathcal{O}(\epsilon_c^{-2})$ and $\mathcal{O}(\epsilon_c^{-1})$ iterations to reach a ϵ_c-stationary feasible point for case I and case II, respectively. The only difference lies in the constant. For example, the inexact algorithm needs roughly $(\tilde{\kappa}_2 - \kappa_2)\times 2(\kappa_c/\nu\sigma\kappa_v)$ more iterations in the worst case. For the first order stationarity measure, both cases yield the same $\mathcal{O}(\epsilon_L^{-4})$ KKT complexity.

6 Numerical Experiments

In this section, we validate the performance of our inexact two-stepsize SQP (ITSQP) method with numerical experiments. We tested ITSQP together with the original stochastic SQP method (SSQP) [3] and a step-lengthening stochastic SQP method (SSQPL) on a subset of the equality constrained optimization problems from CUTEst [18]. Since singularity is detected in more than half of the problems, we use the step decomposition strategy for all three algorithms.

We use an experimental setup similar to [3], with a total of 110 equality constrained problems. Multiple levels of stochasticity (noise follows $\mathcal{N}(0, \epsilon_N I)$, $\epsilon_N \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$) are injected into the gradient estimates to simulate the stochastic gradient. To "tune" the algorithms, we tested 5 different fixed scaling stepsizes $\beta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ under 15 different seeds. Therefore, a total of 440(=110×4) problems are tested, and for each problem, every algorithm is run 75(=5 × 15) times with different stepsizes. For the stepsize α_k, we use the default stepsize choosing scheme in [3] for SSQP and a step-lengthening stepsize for SSQPL.

For each run of the algorithm, we give a budget of 10,000 iterations and report the best iterate with the following scheme: for any iterate k, if $\|c_k\|_\infty \leqslant 10^{-6}$, we treat it as a feasible point. We then pick the best iterate with the lowest KKT error $\|\nabla f_k + J_k^T y_k^{\text{true}}\|_\infty$ from all feasible points. If $\|\nabla f_k + J_k^T y_k^{\text{true}}\|_\infty \leqslant 10^{-4}$, we terminate the algorithm. If such feasible points do not exist, we report the KKT measure of the most feasible point (one with the lowest $\|c_k\|_\infty$). This is commonly used in the stochastic SQP literature, such as [15] and [28].

The figures contain the KKT errors and infeasibility errors of three algorithms tested under different noise levels. For each of the 440 problems, we compared the average infeasibility errors and KKT errors over five different stepsizes, and chose the one stepsize that yields the most best results to plot the figures. It is $\alpha = 1$ for SSQP and SSQPL, and $\beta_k = 10^{-3}$ for ITSQP.

We can conclude from the figures that our ITSQP method outperforms the SSQP and SSQPL methods in terms of the infeasibility measure across all noise levels. This advantage becomes

increasingly pronounced as the noise level increases, highlighting the superior convergence rate of our two-stepsize algorithm.

The SSQP and SSQPL methods compute the merit parameter sequence τ_k and choose the stepsize based on τ_k, so they achieve better KKT convergence. We argue that their lower KKT errors are obtained in the cost of computational resources. The higher KKT errors may also be attributed to the more pessimistic stepsize α_k we use in our algorithm.

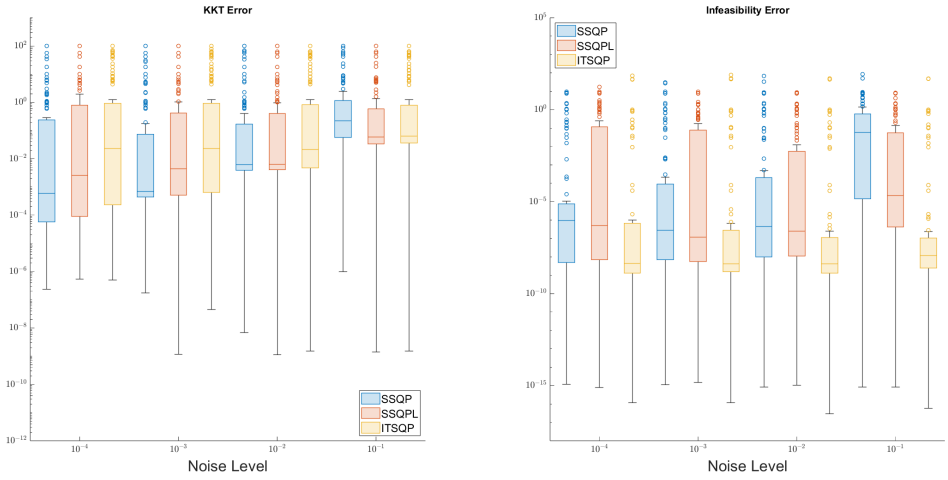


Figure 1: comparison of KKT error and infeasibility error

7 Conclusion

In our paper, we propose and analyze an inexact two-stepsize stochastic SQP method that can handle possible rank deficiency in constraint Jacobians. We prove that our algorithm can converge to a stationary KKT point when the merit parameter sequence is bounded above zero. When the sequence vanishes, the algorithm can converge to an infeasible stationary point, which is the best that one can hope for.

We also prove the first-known $\mathcal{O}(\epsilon_c^{-2})$ complexity of the infeasibility measure $\|J_k^T c_k\|_2$ under mild assumptions, and an improved $\mathcal{O}(\epsilon_c^{-1})$ complexity under the LICQ, which matches existing optimal result. Numerical experiments also show that our algorithm has a better infeasibility convergence rate than those without the two-stepsize scheme. The inexact subproblem solver can also boost the speed of the algorithm.

Future research can be conducted on extending the work to inequality constrained problems and problems with stochastic constraints. For the iterative subproblem solver, whether the scaling parameter in the termination test can be removed needs further attention.

References

[1] Albert S Berahas, Raghu Bollapragada, and Baoyu Zhou. An adaptive sampling sequential quadratic programming method for equality constrained stochastic optimization. *arXiv preprint arXiv:2206.00712*, 2022.

[2] Albert S Berahas, Frank E Curtis, Michael J O'Neill, and Daniel P Robinson. A stochastic sequential quadratic optimization algorithm for nonlinear-equality-constrained optimization with rank-deficient jacobians. *Mathematics of Operations Research*, 49(4):2212–2248, 2024.

[3] Albert S Berahas, Frank E Curtis, Daniel Robinson, and Baoyu Zhou. Sequential quadratic optimization for nonlinear equality constrained stochastic optimization. *SIAM Journal on Optimization*, 31(2):1352–1379, 2021.

[4] Albert S Berahas, Jiahao Shi, Zihong Yi, and Baoyu Zhou. Accelerating stochastic sequential quadratic programming for equality constrained optimization using predictive variance reduction. *Computational Optimization and Applications*, 86(1):79–116, 2023.

[5] Albert S Berahas, Jiahao Shi, and Baoyu Zhou. Optimistic noise-aware sequential quadratic programming for equality constrained optimization with rank-deficient jacobians. *arXiv preprint arXiv:2503.06702*, 2025.

[6] Albert S Berahas, Miaolan Xie, and Baoyu Zhou. A sequential quadratic programming method with high-probability complexity bounds for nonlinear equality-constrained stochastic optimization. *SIAM Journal on Optimization*, 35(1):240–269, 2025.

[7] Dimitri Bertsekas. *Network optimization: continuous and discrete models*, volume 8. Athena Scientific, 1998.

[8] John T Betts. *Practical methods for optimal control and estimation using nonlinear programming*. SIAM, 2010.

[9] Raghu Bollapragada, Cem Karamanli, Brendan Keith, Boyan Lazarov, Socratis Petrides, and Jingyi Wang. An adaptive sampling augmented lagrangian method for stochastic optimization with deterministic constraints. *Computers & Mathematics with Applications*, 149:239–258, 2023.

[10] Digvijay Boob, Qi Deng, and Guanghui Lan. Level constrained first order methods for function constrained optimization. *Mathematical Programming*, 209(1):1–61, 2025.

[11] Richard H Byrd, Jean Charles Gilbert, and Jorge Nocedal. A trust region method based on interior point techniques for nonlinear programming. *Mathematical programming*, 89(1):149–185, 2000.

[12] Hao Chen, Gonzalo E Constante Flores, and Can Li. Physics-informed neural networks with hard linear equality constraints. *Computers & Chemical Engineering*, 189:108764, 2024.

[13] Frank E Curtis, Jorge Nocedal, and Andreas Wächter. A matrix-free algorithm for equality constrained optimization problems with rank-deficient jacobians. *SIAM Journal on Optimization*, 20(3):1224–1249, 2010.

[14] Frank E Curtis, Michael J O'Neill, and Daniel P Robinson. Worst-case complexity of an sqp method for nonlinear equality constrained stochastic optimization. *Mathematical Programming*, 205(1):431–483, 2024.

[15] Frank E Curtis, Daniel P Robinson, and Baoyu Zhou. A stochastic inexact sequential quadratic optimization algorithm for nonlinear equality-constrained optimization. *INFORMS Journal on Optimization*, 6(3-4):173–195, 2024.

[16] Yuchen Fang, Sen Na, Michael W Mahoney, and Mladen Kolar. Fully stochastic trust-region sequential quadratic programming for equality-constrained optimization problems. *SIAM Journal on Optimization*, 34(2):2007–2037, 2024.

[17] Saeed Ghadimi and Guanghui Lan. Stochastic first-and zeroth-order methods for nonconvex stochastic programming. *SIAM journal on optimization*, 23(4):2341–2368, 2013.

[18] Serge Gratton and Ph L Toint. S2mpj and cutest optimization problems for matlab, python and julia. *Optimization Methods and Software*, pages 1–33, 2025.

[19] Zhanhong Jiang, Chao Liu, Young M Lee, Chinmay Hegde, Soumik Sarkar, and Dongxiang Jiang. The stochastic augmented lagrangian method for domain adaptation. *Knowledge-Based Systems*, 235:107593, 2022.

[20] Lingzi Jin and Xiao Wang. A stochastic primal-dual method for a class of nonconvex constrained optimization. *Computational Optimization and Applications*, 83(1):143–180, 2022.

[21] Drew P Kouri and Denis Ridzal. Inexact trust-region methods for pde-constrained optimization. In *Frontiers in PDE-constrained optimization*, pages 83–121. Springer, 2018.

[22] Zichong Li, Pin-Yu Chen, Sijia Liu, Songtao Lu, and Yangyang Xu. Stochastic inexact augmented lagrangian method for nonconvex expectation constrained optimization. *Computational Optimization and Applications*, 87(1):117–147, 2024.

[23] Qihang Lin, Runchao Ma, and Yangyang Xu. Complexity of an inexact proximal-point penalty method for constrained smooth non-convex optimization. *Computational optimization and applications*, 82(1):175–224, 2022.

[24] Lu Lu, Raphael Pestourie, Wenjie Yao, Zhicheng Wang, Francesc Verdugo, and Steven G Johnson. Physics-informed neural networks with hard constraints for inverse design. *SIAM Journal on Scientific Computing*, 43(6):B1105–B1132, 2021.

[25] Sen Na, Mihai Anitescu, and Mladen Kolar. An adaptive stochastic sequential quadratic programming with differentiable exact augmented lagrangians. *Mathematical Programming*, 199(1):721–791, 2023.

[26] Jorge Nocedal and Stephen J Wright. *Numerical optimization*. Springer, 2006.

[27] Emmanuel Omotayo Omojokun. *Trust region algorithms for optimization with nonlinear equality and inequality constraints*. University of Colorado at Boulder, 1989.

[28] Michael J O'Neill. A two stepsize sqp method for nonlinear equality constrained stochastic optimization. *arXiv preprint arXiv:2408.16656*, 2024.

[29] Christopher C Paige and Michael A Saunders. Solution of sparse indefinite systems of linear equations. *SIAM journal on numerical analysis*, 12(4):617–629, 1975.

[30] Haoming Shen, Yang Zeng, and Baoyu Zhou. Sequential quadratic optimization for solving expectation equality constrained stochastic optimization problems. *arXiv preprint arXiv:2503.09490*, 2025.

[31] Qiankun Shi, Xiao Wang, and Hao Wang. A momentum-based linearized augmented lagrangian method for nonconvex constrained stochastic optimization. *Mathematics of Operations Research*, 2025.

[32] Shigeng Sun and Jorge Nocedal. A trust-region algorithm for noisy equality constrained optimization. *arXiv preprint arXiv:2411.02665*, 2024.

[33] Kok Lay Teo, Bin Li, Changjun Yu, Volker Rehbock, et al. Applied and computational optimal control. *Optimization and Its Applications*, 2021.

[34] Robert B Wilson. A simplicial method for convex programming. *Harvard University: Cambridge, MA, USA*, 1963.

[35] Yangyang Xu. Primal-dual stochastic gradient method for convex programs with many functional constraints. *SIAM Journal on Optimization*, 30(2):1664–1692, 2020.

[36] Jingzhao Zhang, Hongzhou Lin, Stefanie Jegelka, Suvrit Sra, and Ali Jadbabaie. Complexity of finding stationary points of nonconvex nonsmooth functions. In *International Conference on Machine Learning*, pages 11173–11182. PMLR, 2020.